REENA MEHTA COLLEGE

of Commerce and Management Studies

ISO 9001-2015 CERTIFIED

Reena Mehta College, Opp Maxus Mall, Bhayander (West),
Thane – 401101

"DATA VISUALIZATION"
CERTIFICATE

The Practical's duly signed in this journal represent the bonafide work by, _____*in SEMESTER V of THIRD Year in Bachelor of Science in Data Science (BSc.DS) Roll No.____. In the COMPUTER LABORATORY of this College during the session of 2026-27.*

_____ _____

LECTURER IN CHARGE **HEAD OF DEPARTMENT**

(BSC.DATA SCIENCE)

_____ _____

EXTERNAL EXAMINER **PRINCIPAL**

INDEX

Sr. No.	Name of Practicals	Date of Experiment	Date of Submission	Remarks
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2	Working with Data Sources and Data Preparation in Tableau			
3	Creating Advanced Visualizations in Tableau			
4	Performing Data Analysis using Calculations in Tableau			
5	Analyzing Data using Table Calculations in Tableau			
6	Designing Effective and Visually Appealing Dashboards			
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8	Performing Advanced Analytics using Trends, Clustering, and Forecasting			
9	Cleaning and Structuring Data for Analysis			
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11	Implementing Advanced Visualization Techniques and Customizations			
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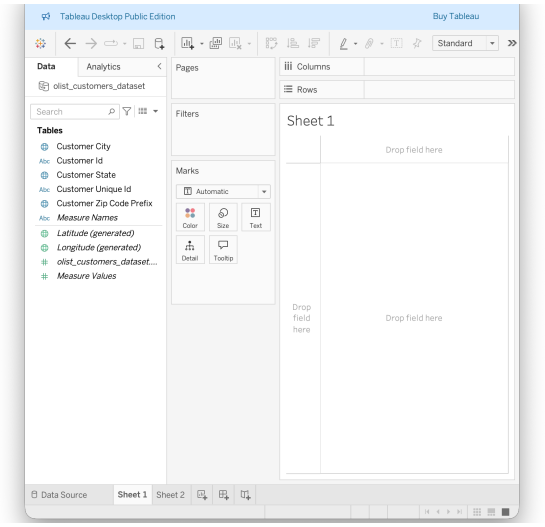
15	Sales Forecasting and Trend Analysis			

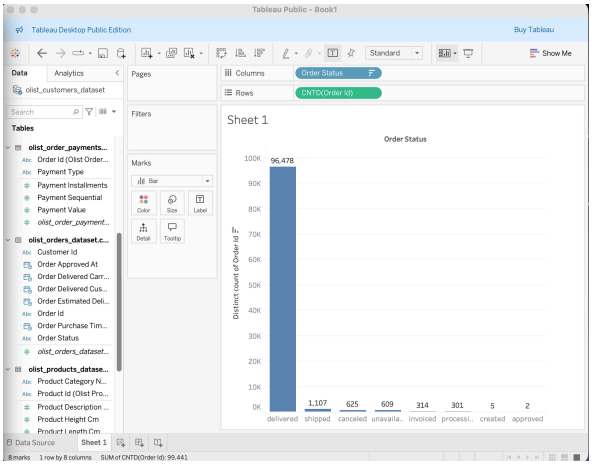
Practical 1

Getting Started with Tableau and Building Basic Visualizations

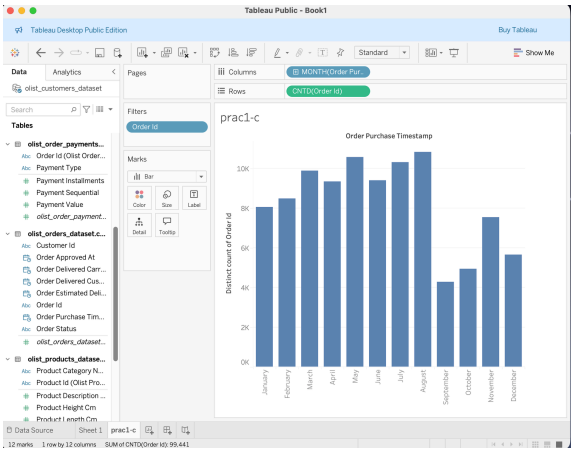
a) Connect to a CSV dataset and explore the Tableau interface

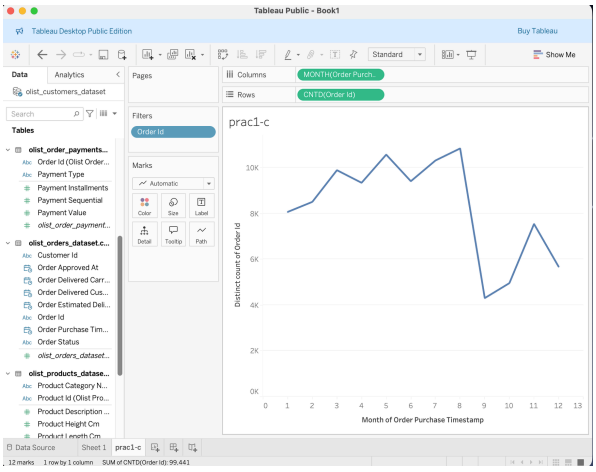


b) Identify and differentiate Measures and Dimensions

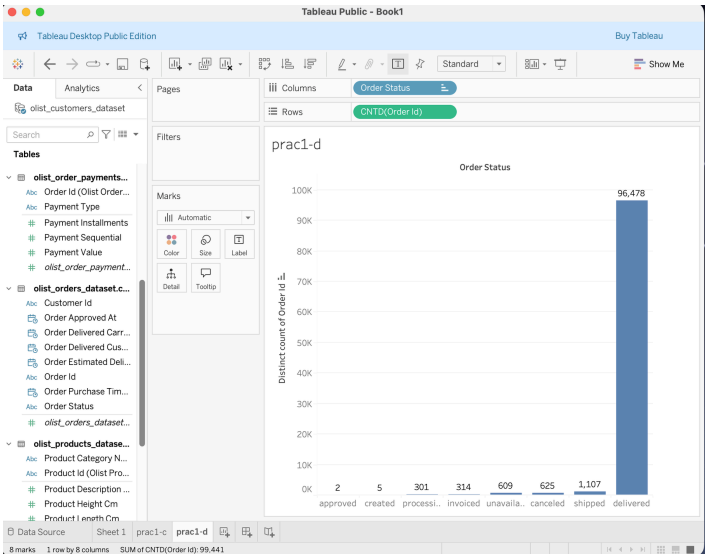


c) Convert fields between discrete and continuous and observe changes

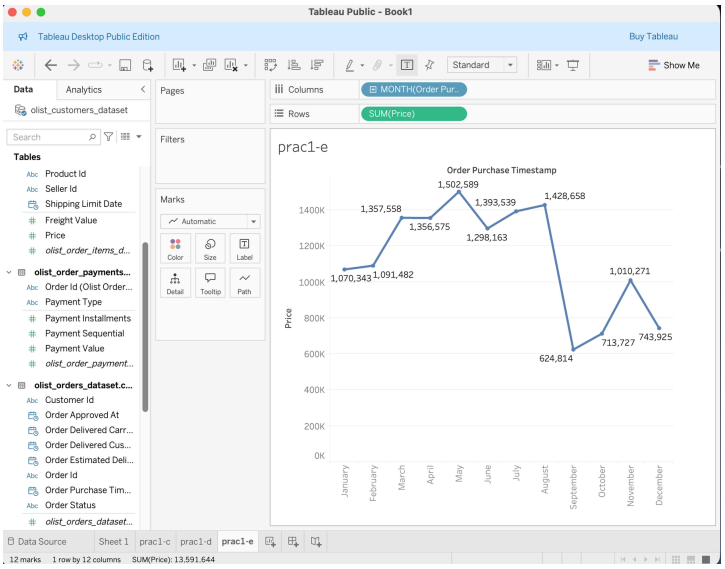




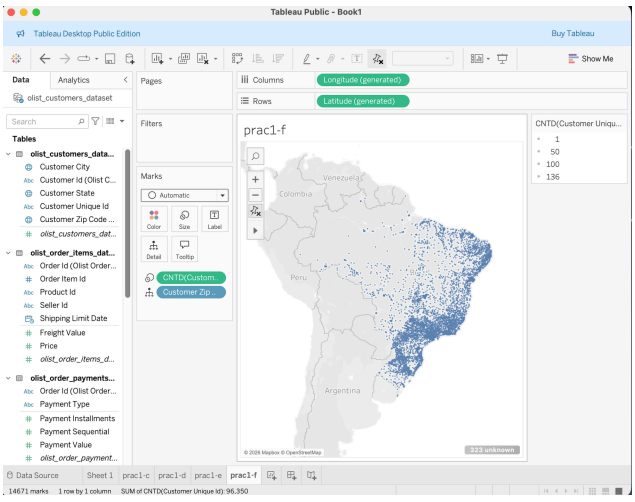
d) Create and customize a basic bar chart



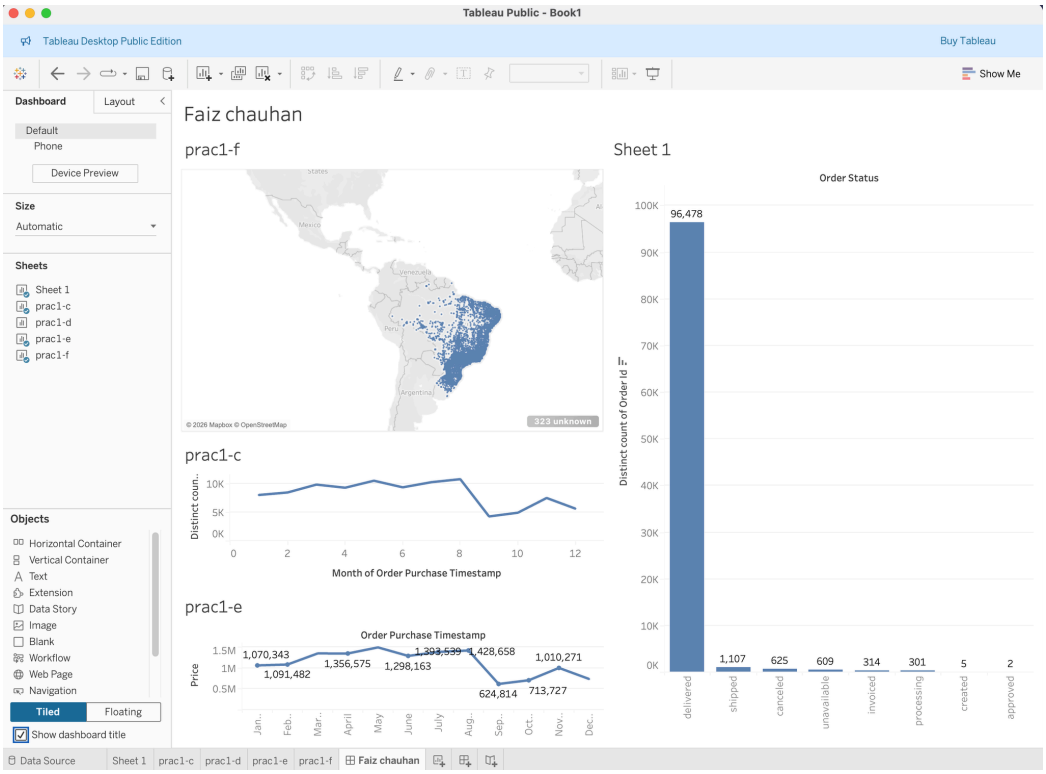
e) Build a line chart for time-series analysis



f) Create a geographic map visualization



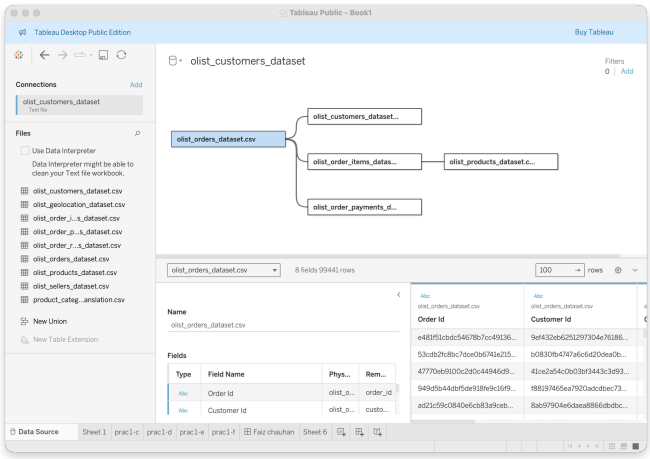
g) Combine multiple charts into a simple interactive dashboard



Practical 2

Working with Data Sources and Data Preparation in Tableau

a) Connect to different data sources (Excel, CSV, Database)



b) Create and use data extracts vs live connections

1. Live Connection: A live connection means Tableau directly connects to the original data source.

Advantages

- Data can be more up-to-date.
- No separate extract needs to be maintained.

Disadvantages

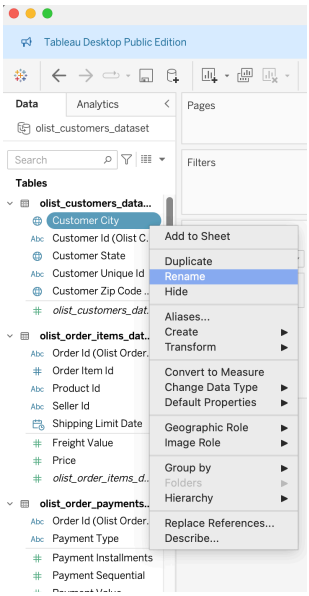
- Performance depends on the database.
- Large or complex queries may be slower.

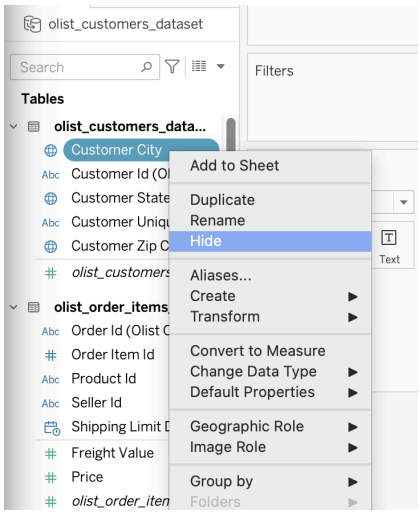
2. Extract Connection: An extract creates a stored copy of the required data that Tableau can query.

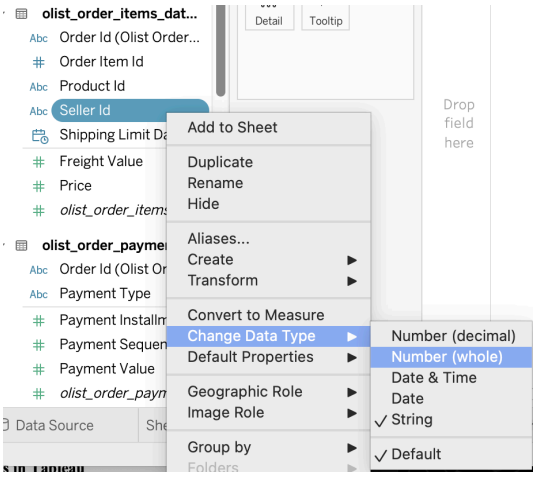
Advantages

- Usually faster for analysis.
- Can reduce load on the original database.

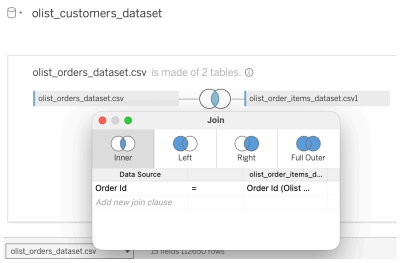
c) Manage metadata (rename, hide, change data types)

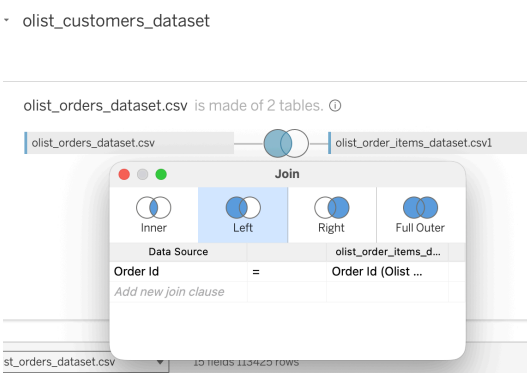


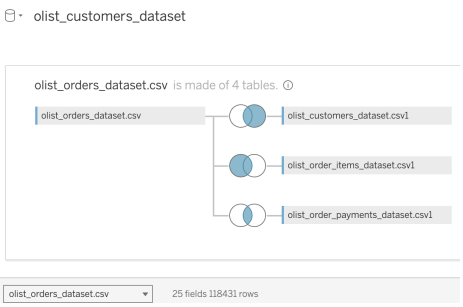




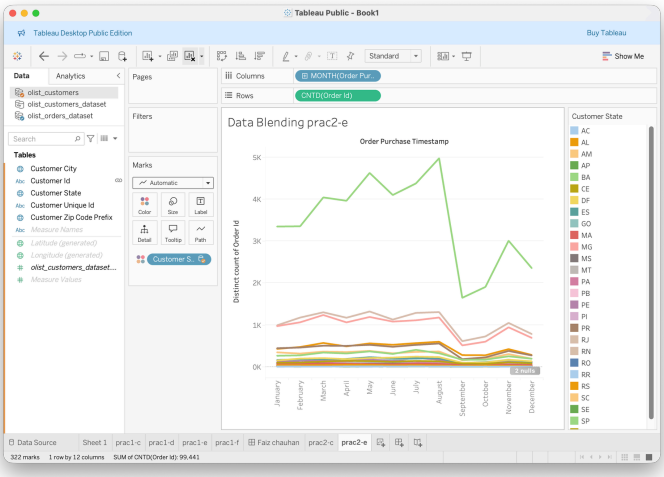
d) Perform joins (inner, left, right) on multiple tables



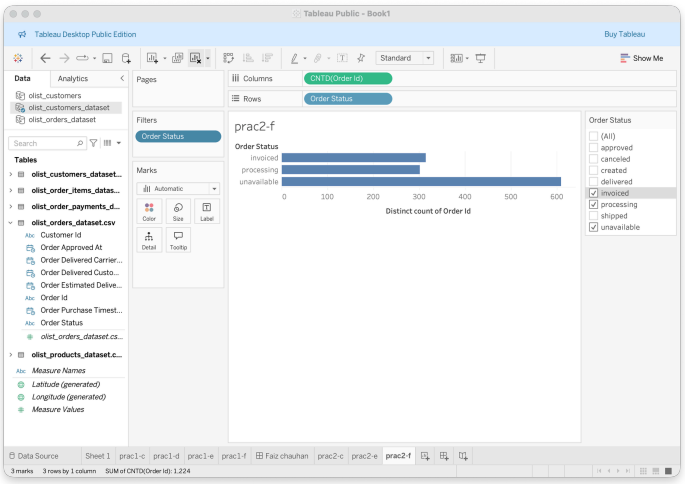


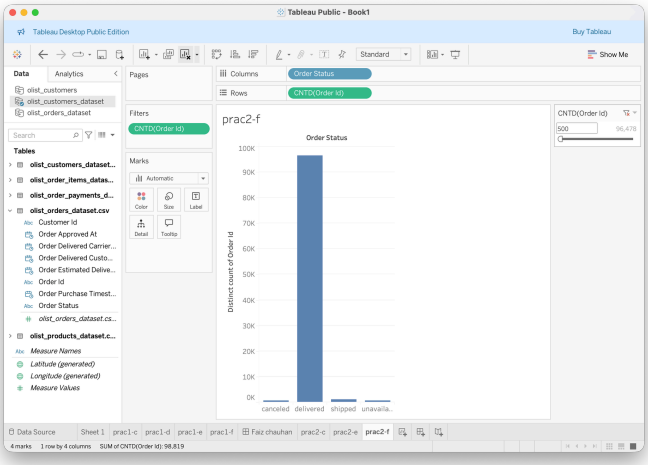


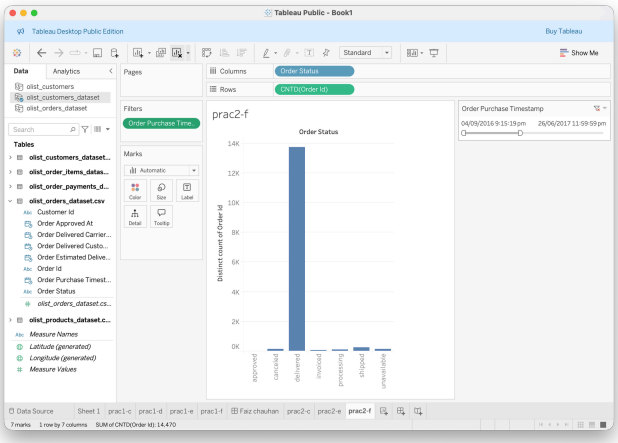
e) Apply data blending between two sources



f) Implement different types of filters (dimension, measure, date)

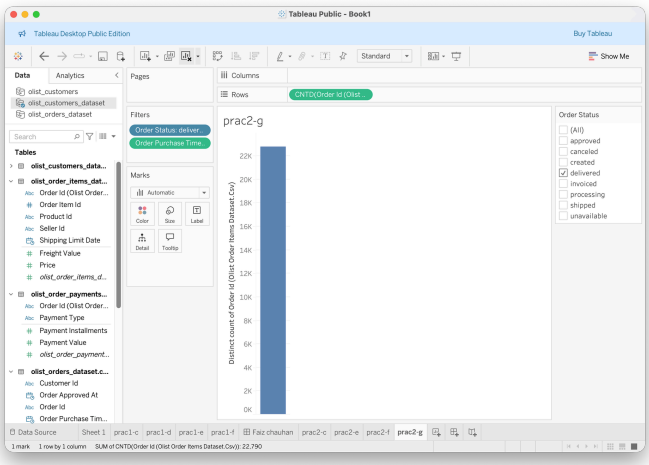


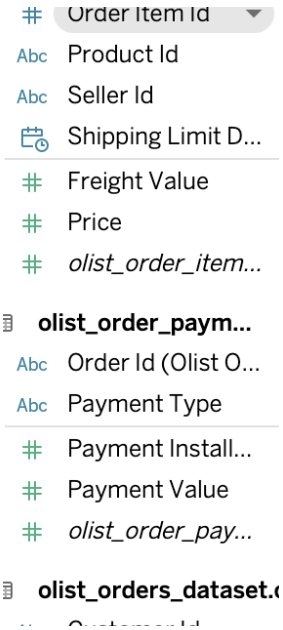




g) Optimize performance using extracts and filtering

Hiding unused fields and filtering by columns to improve performance during Analysis and reducing workload on system

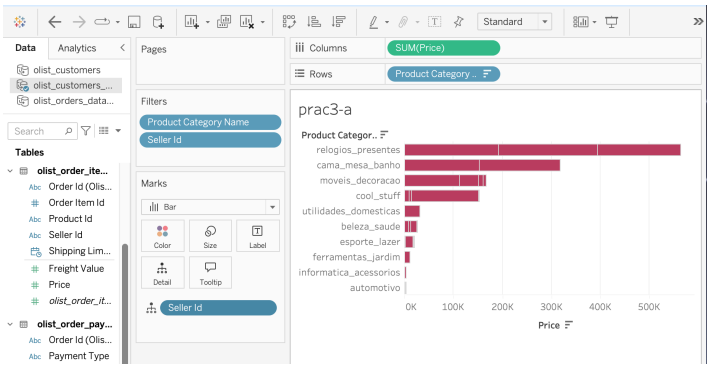


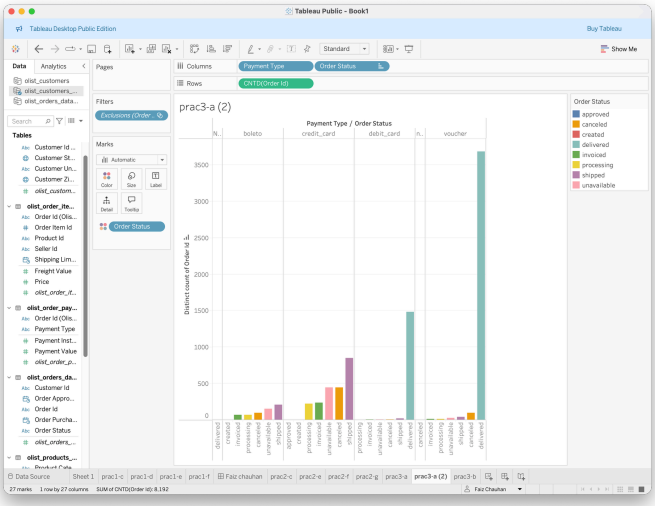


Practical 3

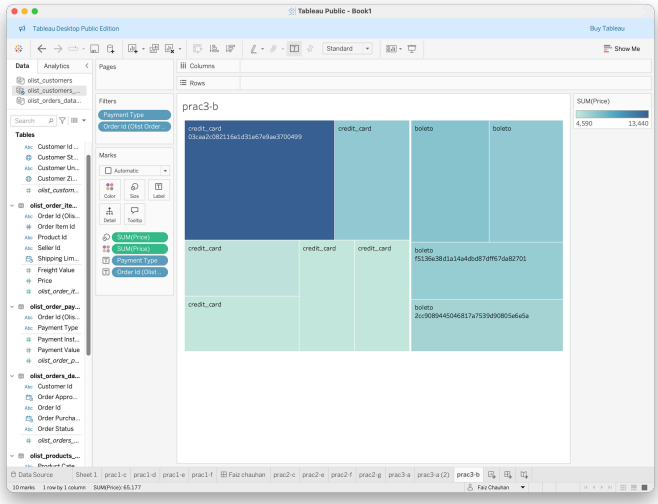
Creating Advanced Visualizations in Tableau

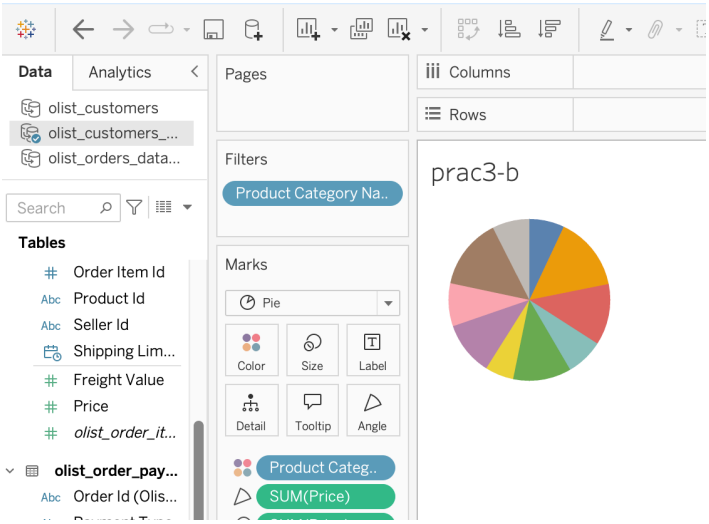
a) Create stacked and grouped bar charts for comparison





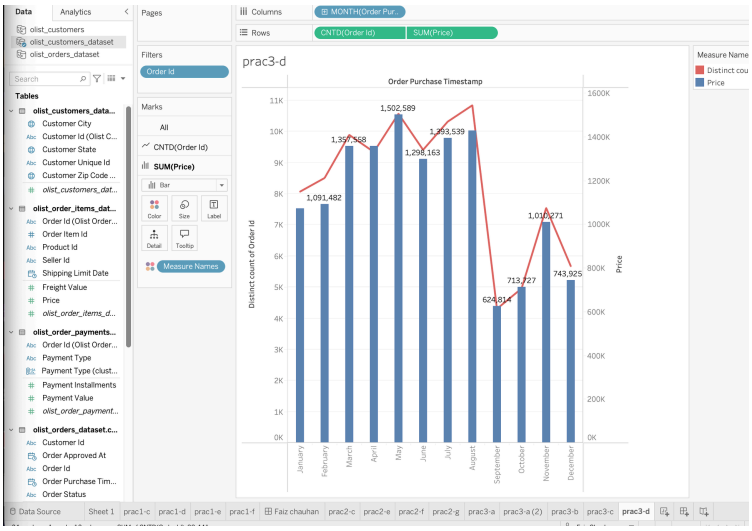
b) Build treemaps and pie charts for part-to-whole analysis



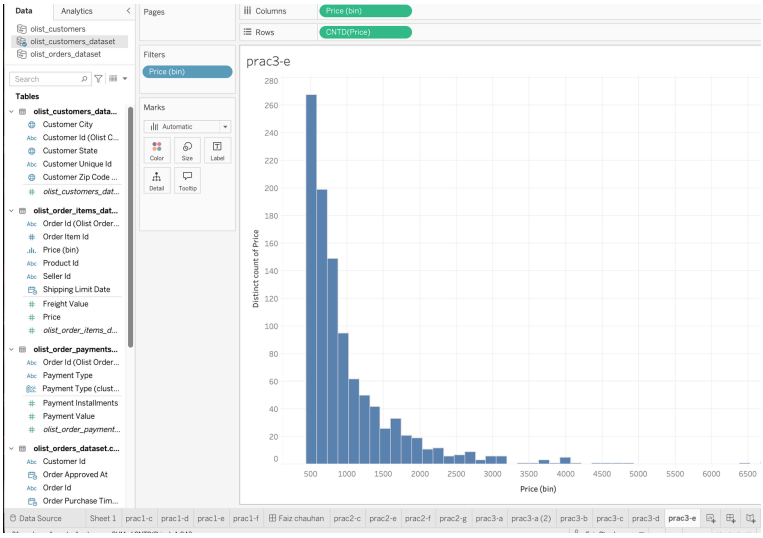


c) Design scatter plots to identify relationships

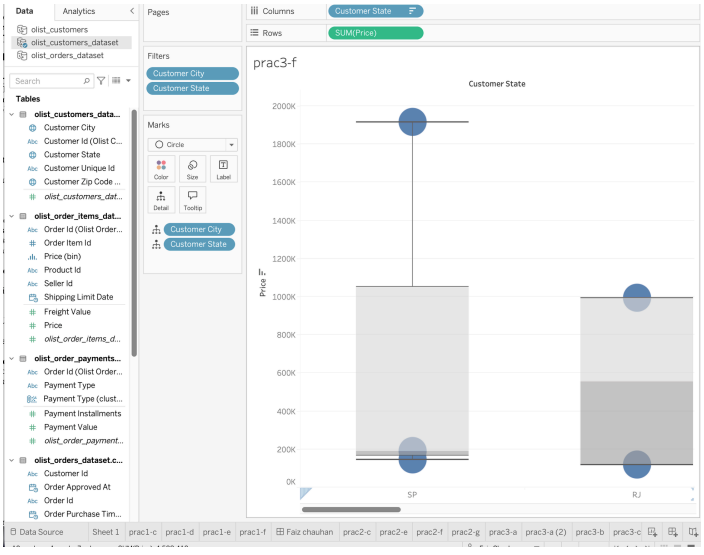
d) Create dual-axis and combination charts



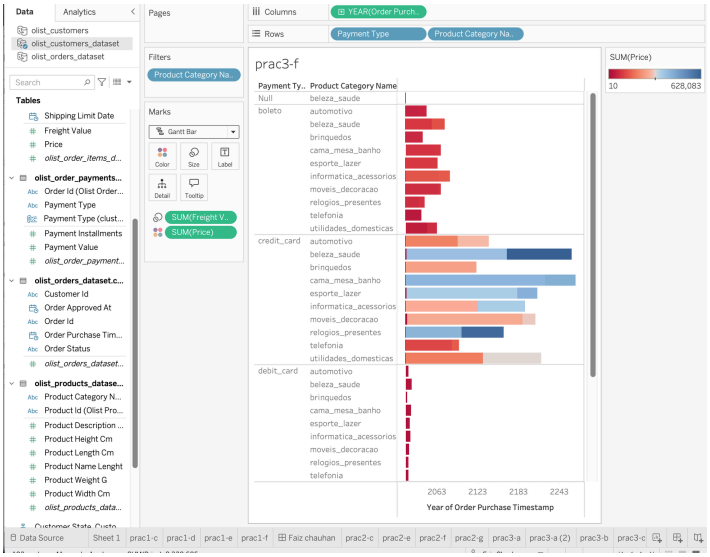
e) Develop histograms for distribution analysis



f) Build box-and-whisker plots



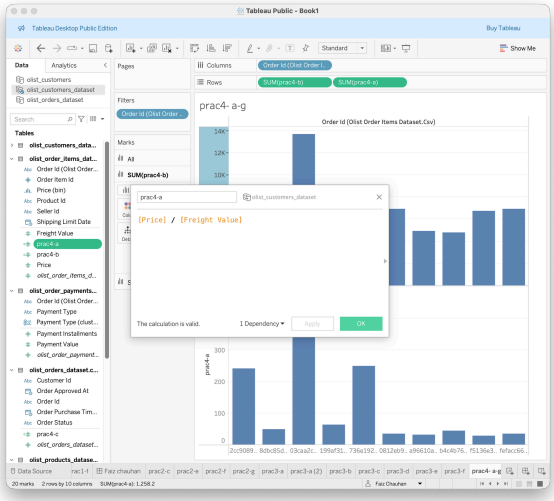
g) Use Gantt charts and advanced date visualizations



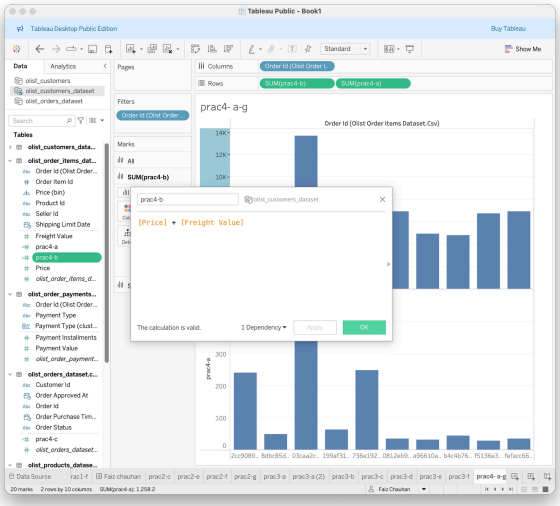
Practical 4

Performing Data Analysis using Calculations in Tableau

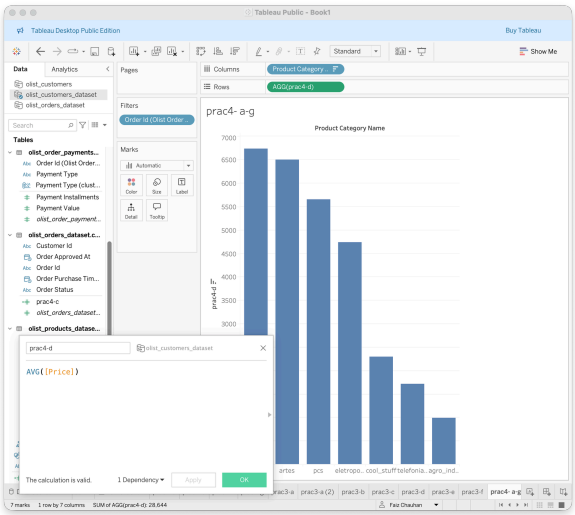
a) Create basic calculated fields using formulas



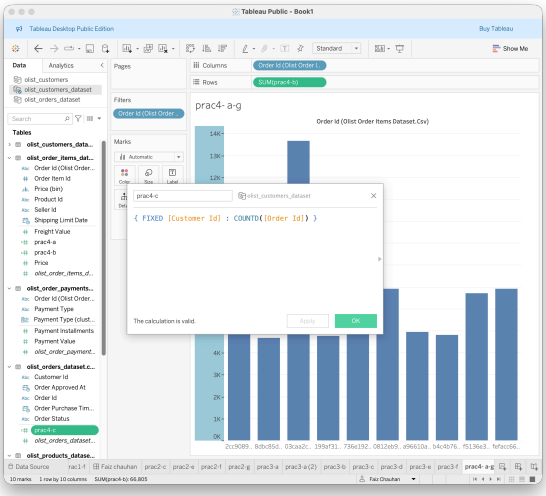
b) Perform row-level calculations



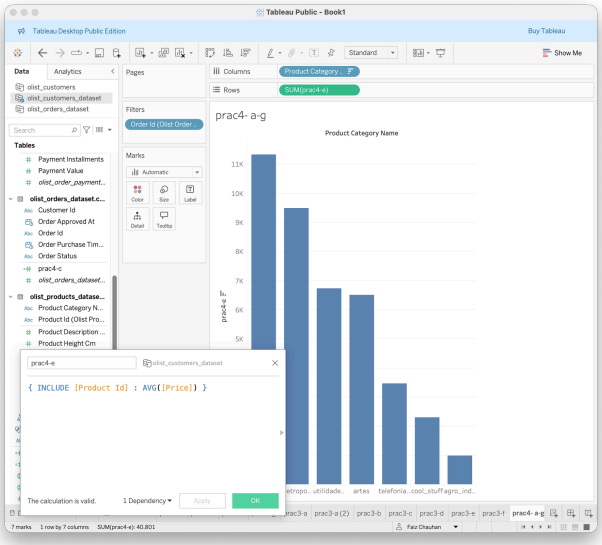
c) Perform aggregate-level calculations



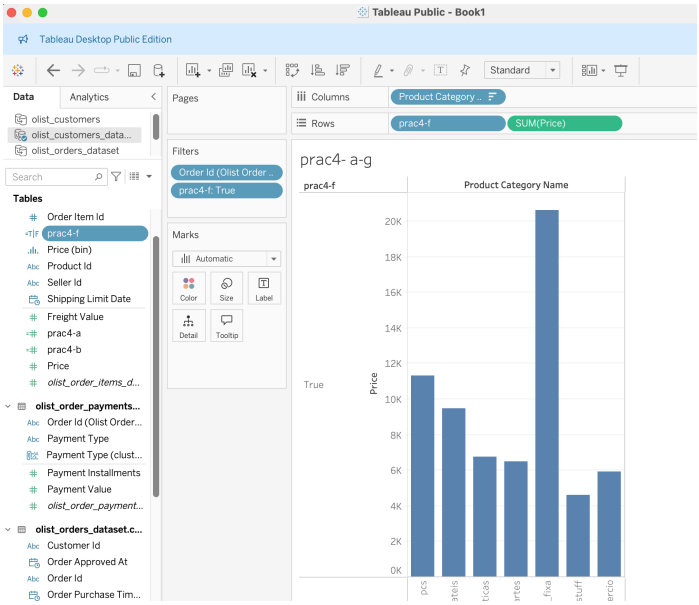
d) Implement FIXED Level of Detail (LOD) calculations



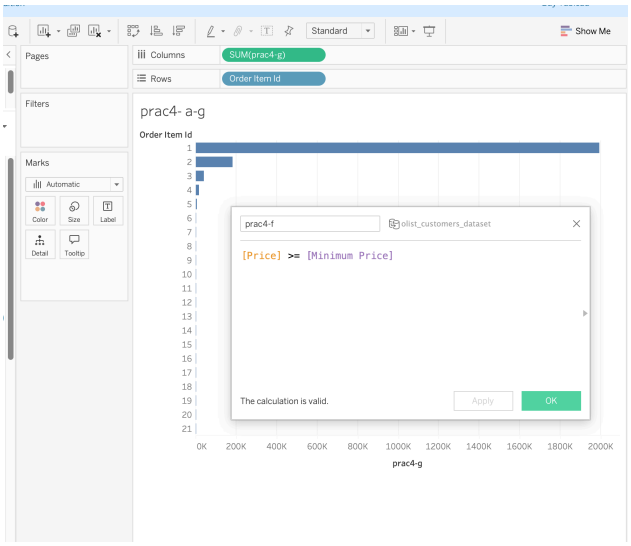
e) Use INCLUDE and EXCLUDE LOD expressions



f) Create and use parameters for dynamic analysis



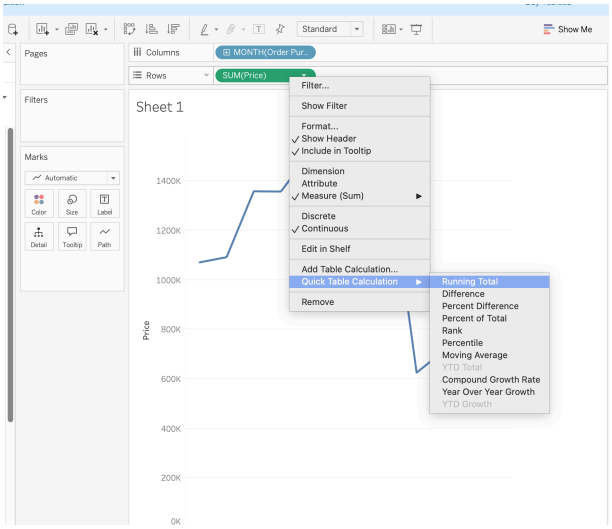
g) Apply calculations to fix data issues and enhance insights

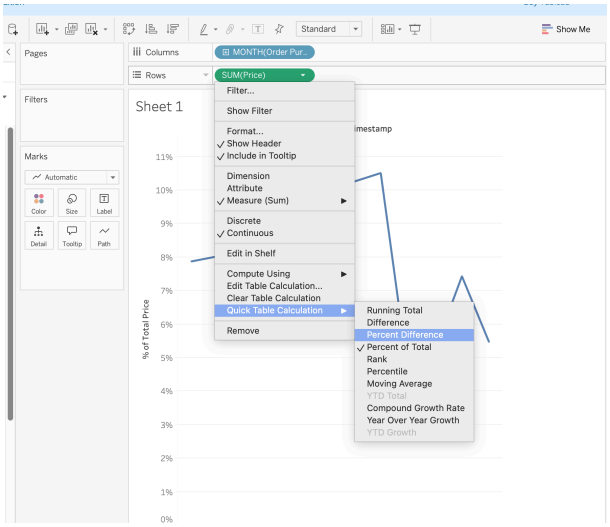


Practical 5

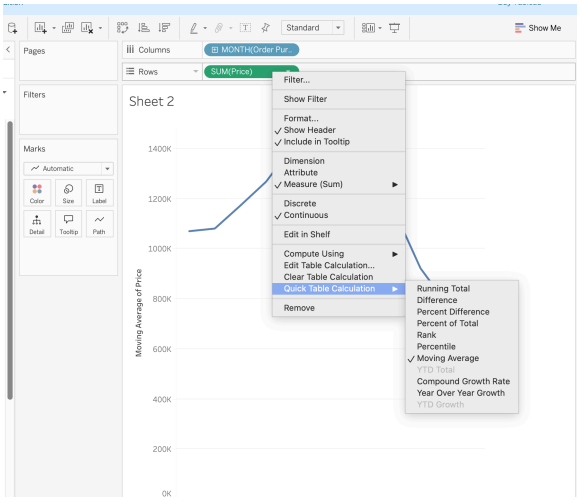
Analyzing Data using Table Calculations in Tableau

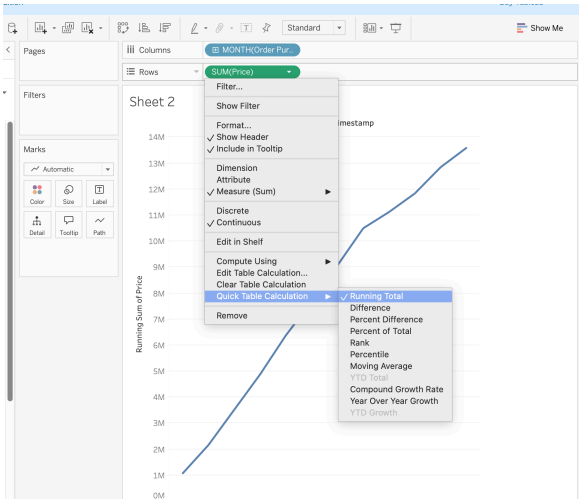
a) Apply quick table calculations (running total, percent of total)



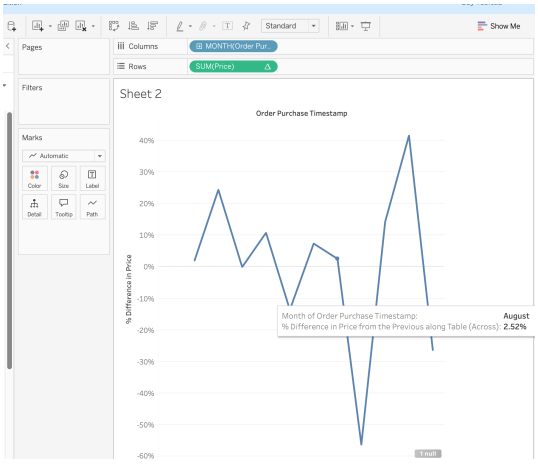


b) Create moving averages and cumulative metrics

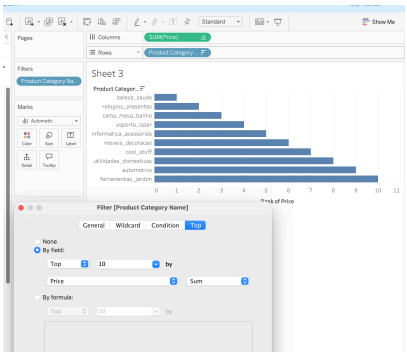


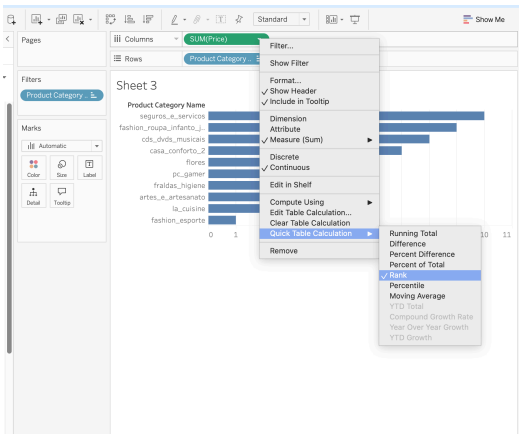


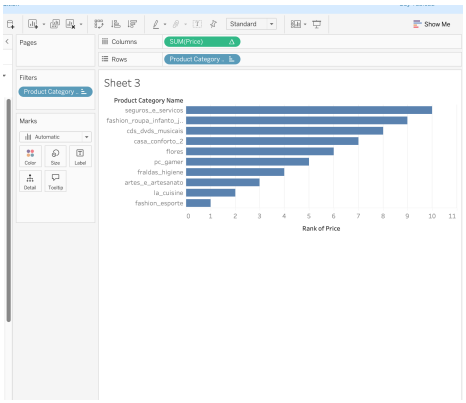
c) Perform year-over-year growth analysis



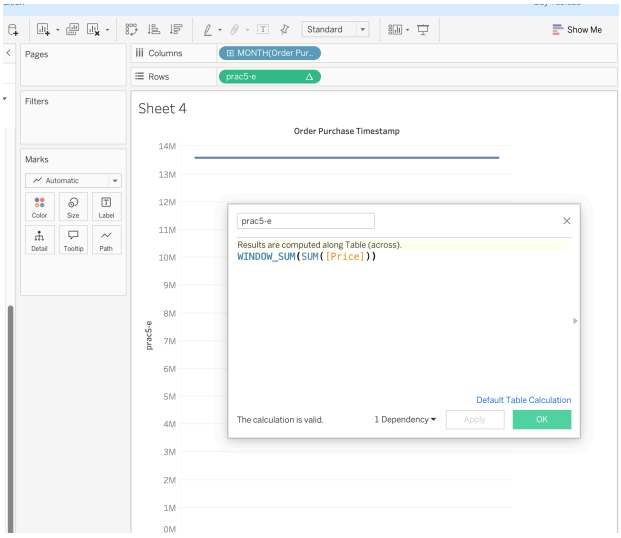
d) Use ranking functions (Top N, Bottom N)

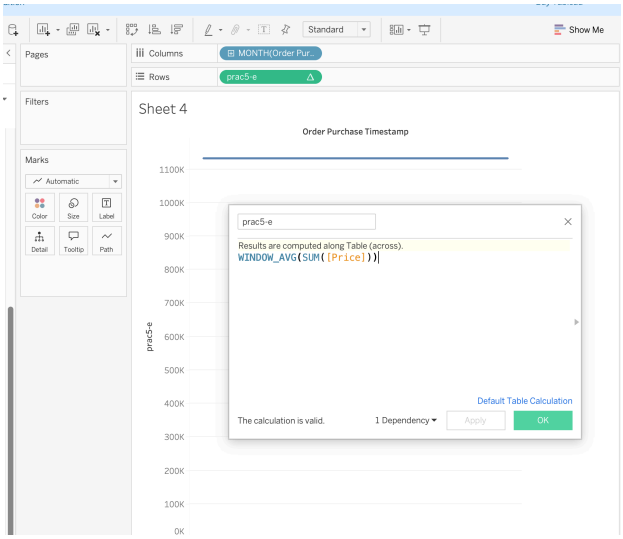




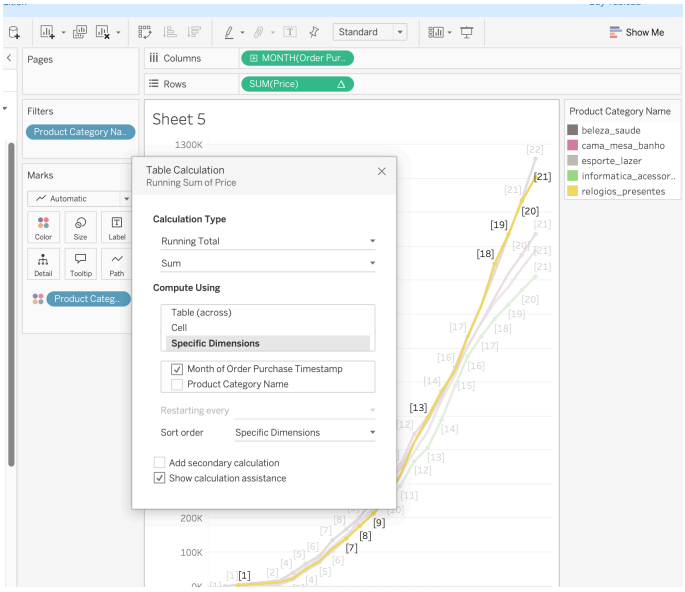


e) Implement window functions (WINDOW_SUM, WINDOW_AVG)

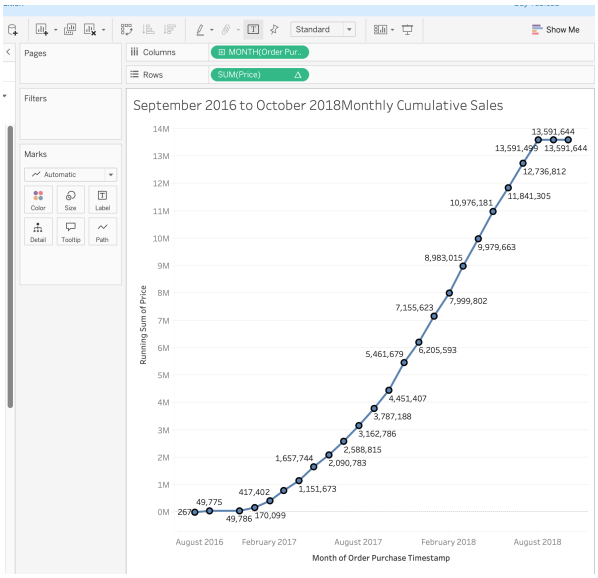




f) Customize addressing and partitioning



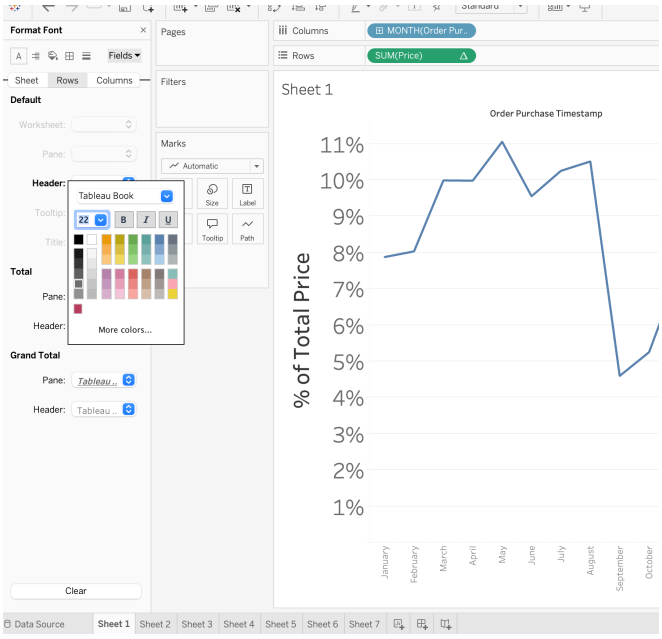
g) Create dynamic titles and labels using table calculations

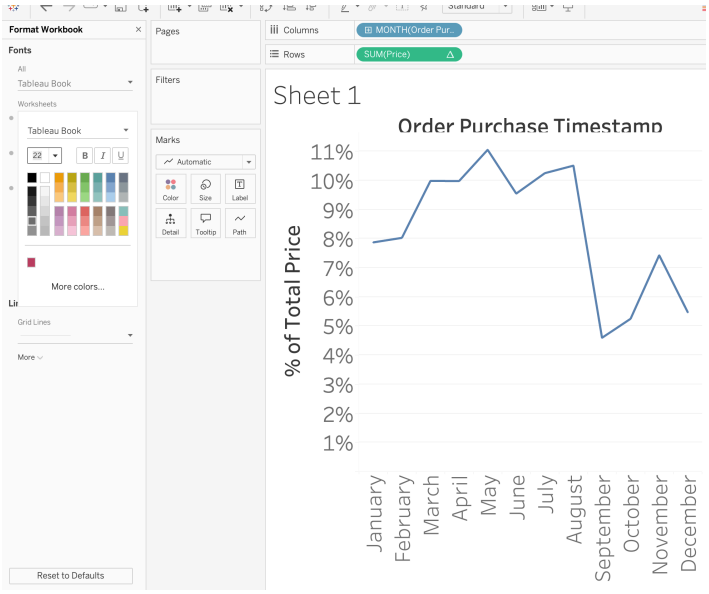


Practical 6

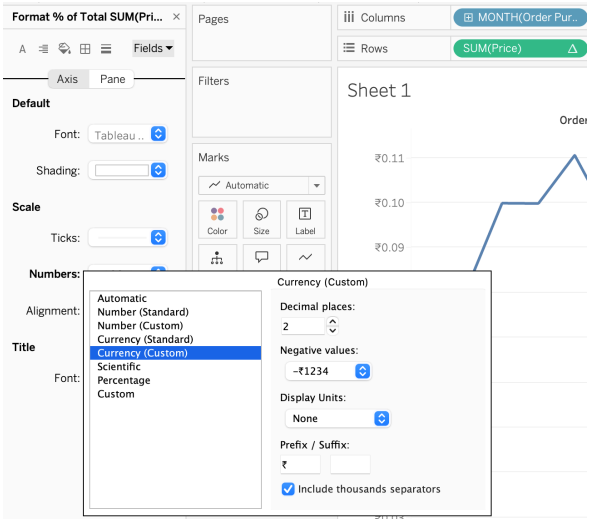
Designing Effective and Visually Appealing Dashboards

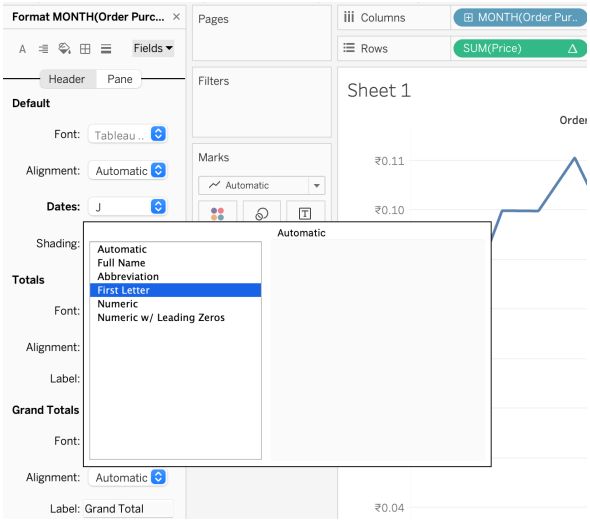
a) Apply worksheet-level and workbook-level formatting



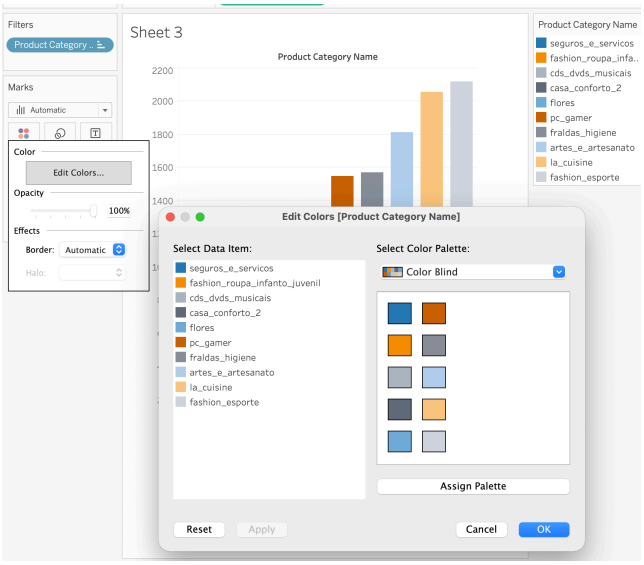


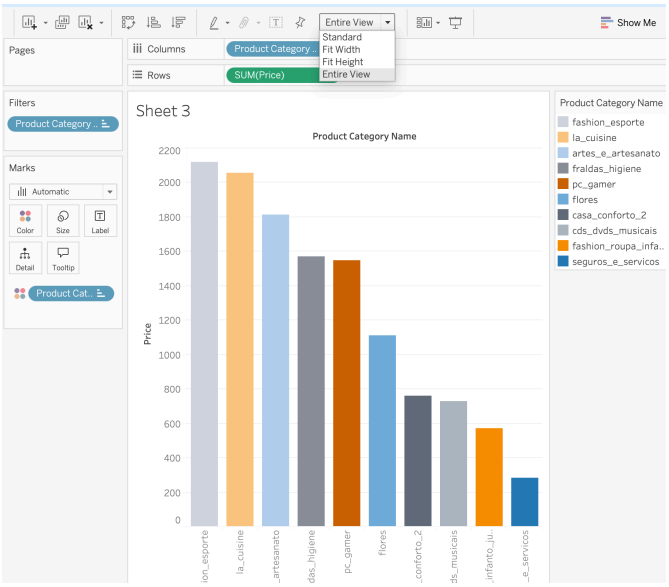
b) Customize number, date, and null value formatting



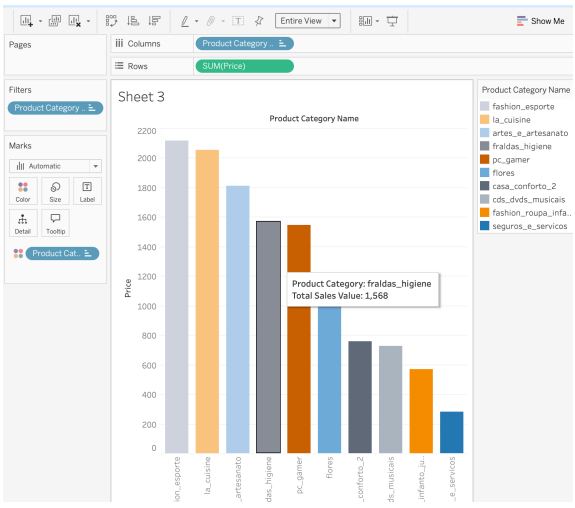


c) Design effective color schemes and layouts

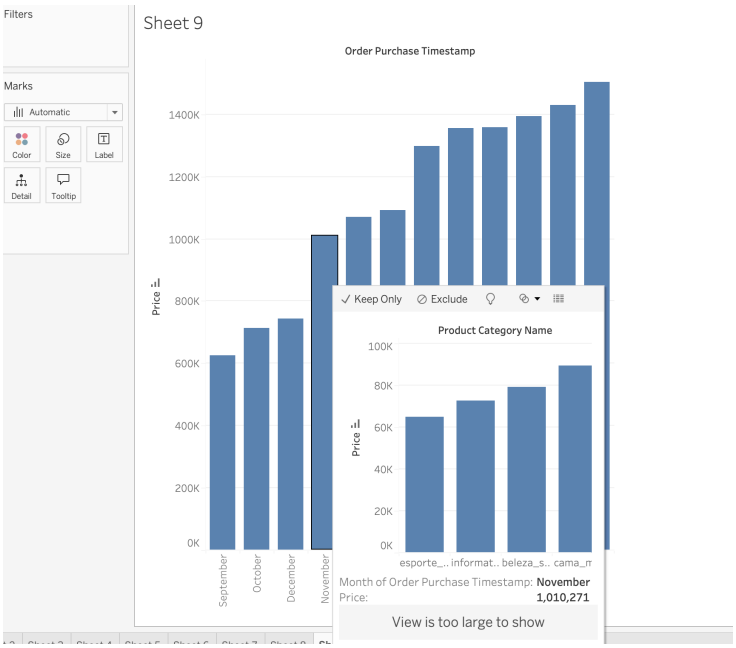




d) Create informative tooltips



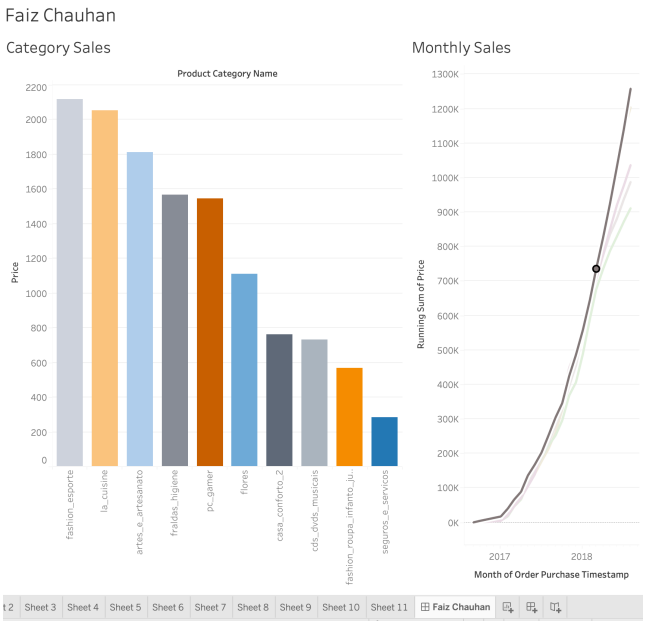
e) Implement “Viz in Tooltip”



f) Improve readability and user experience

Improvements in Readability and User Experience

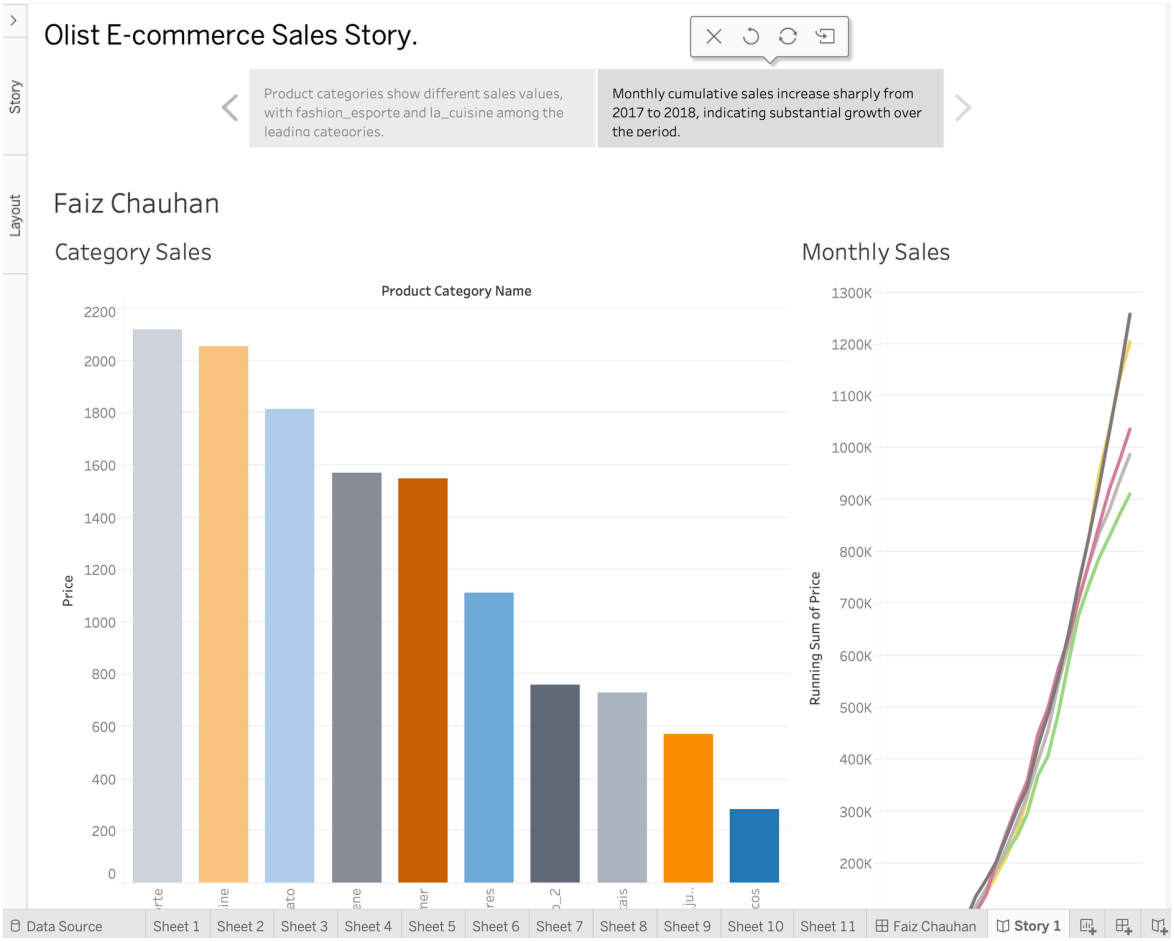
- **Arranged the charts in a clear and organized dashboard layout.**
- **Added a descriptive dashboard title to explain the purpose of the analysis.**
- **Used consistent colours and formatting to make the visualizations easier to understand.**
- **Removed unnecessary legends and visual clutter.**
- **Improved chart titles, axis labels, and spacing for better readability.**
- **Set the dashboard size to Automatic so the visualizations fit the available screen space.**
- **Enabled interactive filters to allow users to explore the data more easily.**



g) Apply best practices for visual storytelling

Storytelling with Data

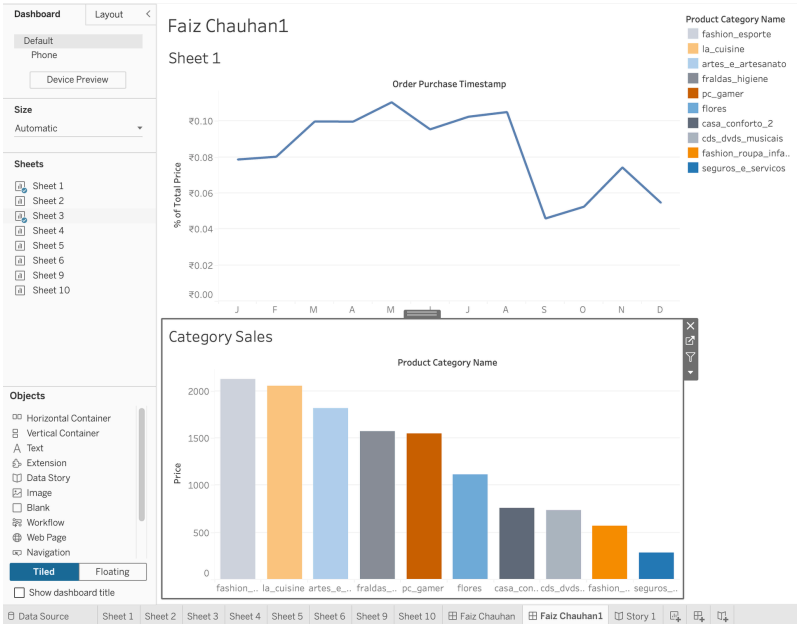
- **Created a Tableau Story to present the sales analysis in a logical sequence.**
- **Used category-wise sales to identify leading product categories.**
- **Presented cumulative sales trends over time to highlight overall growth.**
- **Added captions to explain the key insight from each story point.**
- **Combined visualizations and explanatory text to communicate findings clearly and support data-driven interpretation.**



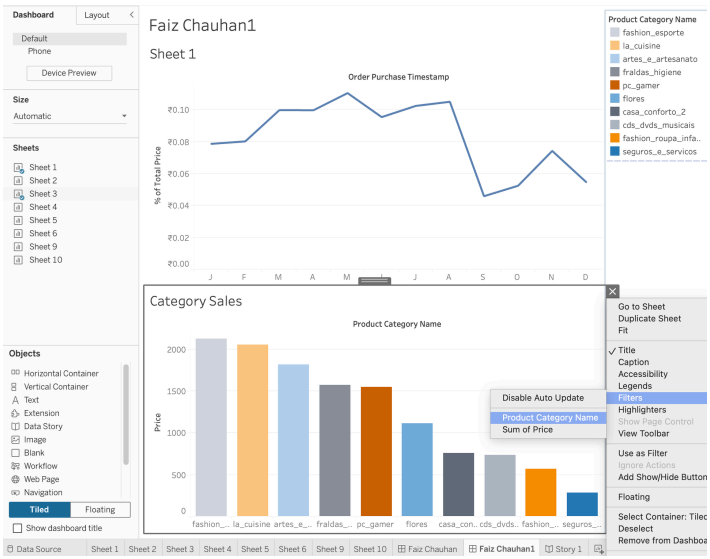
Practical 7

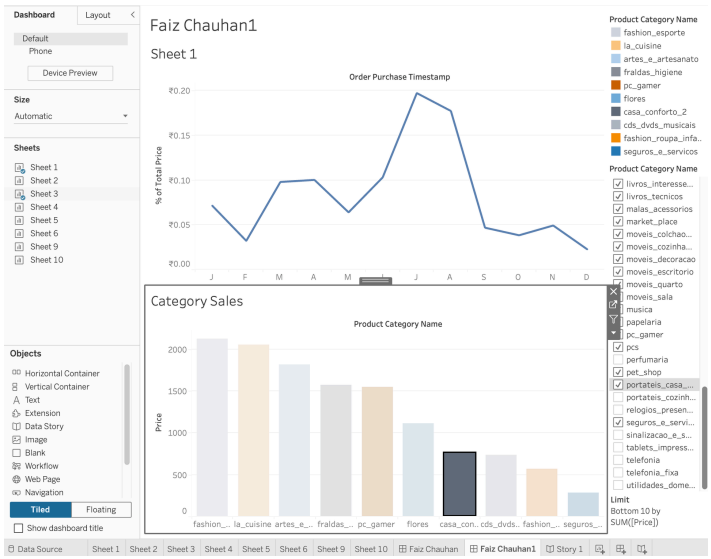
Creating Interactive Dashboards and Data Stories

a) Design a basic dashboard layout using sheets

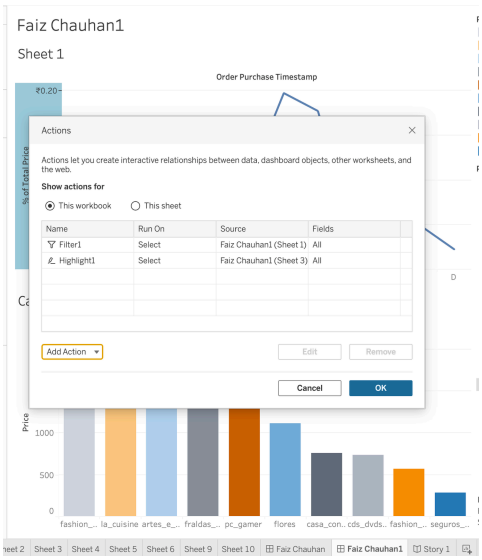


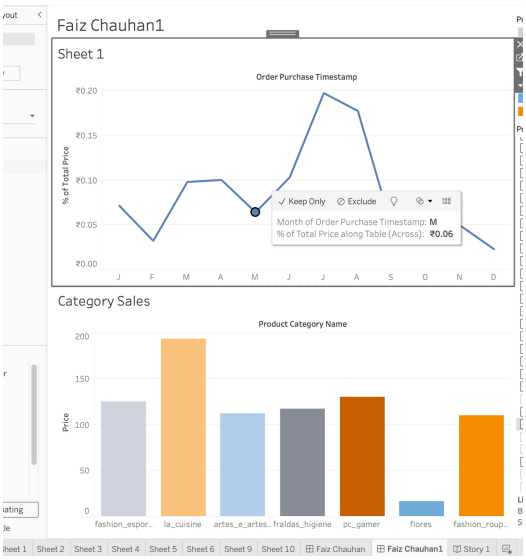
b) Add filters and interactive controls



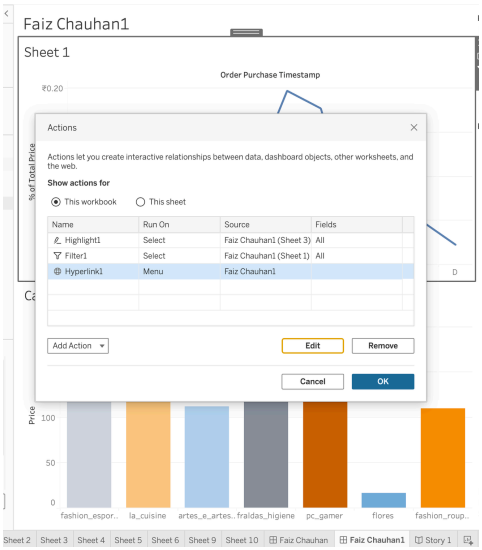


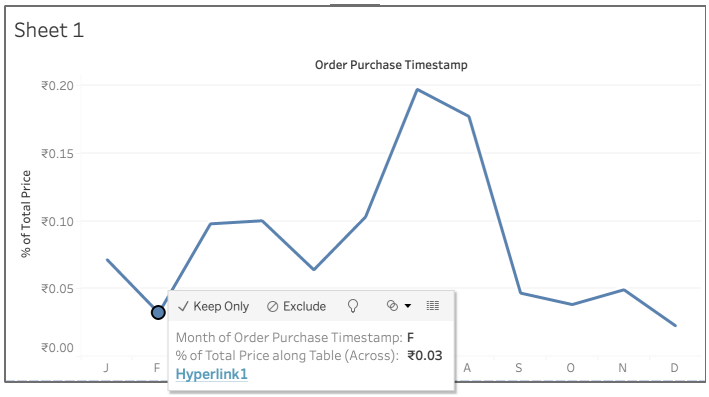
c) Implement highlight and filter actions

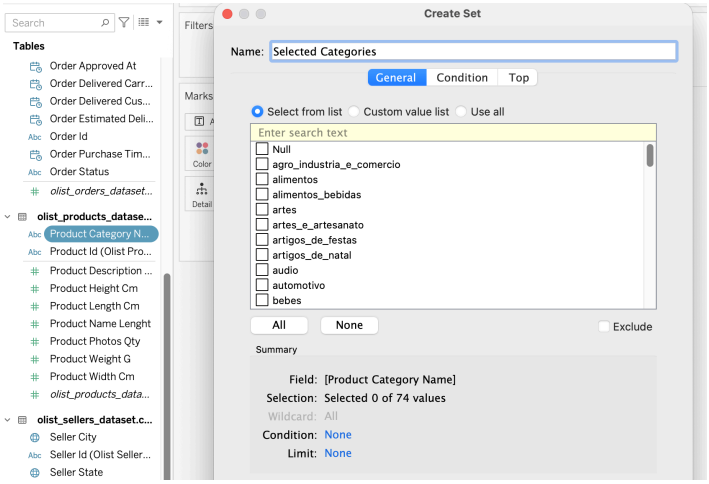


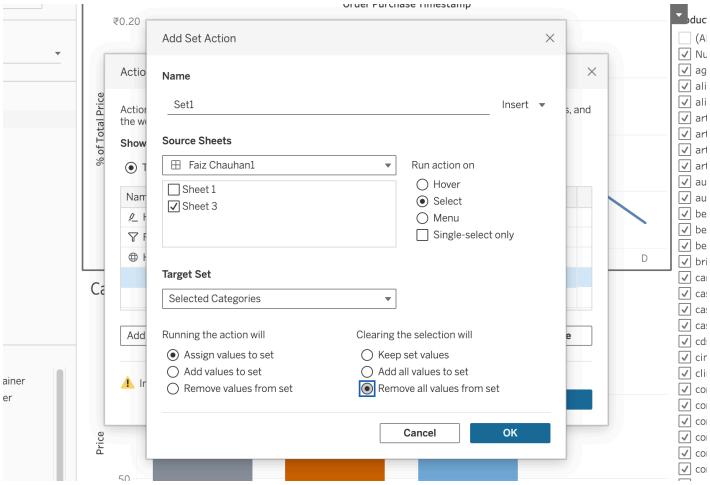


(d) Use URL and set actions for interactivity

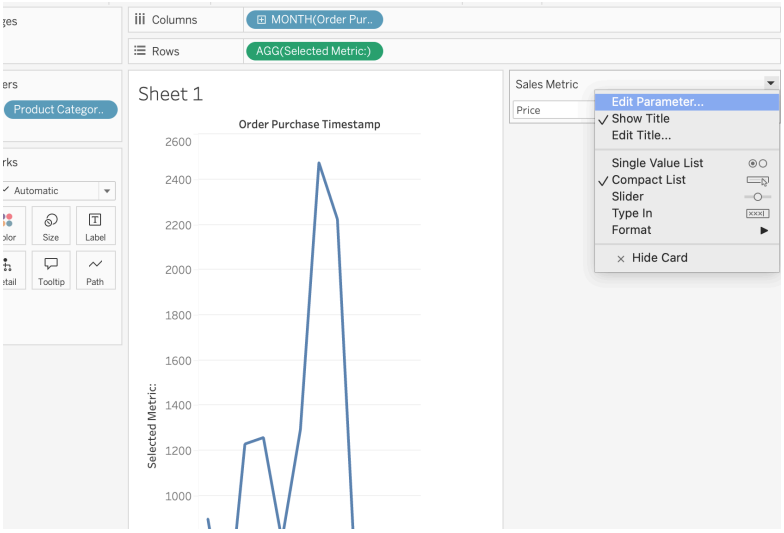




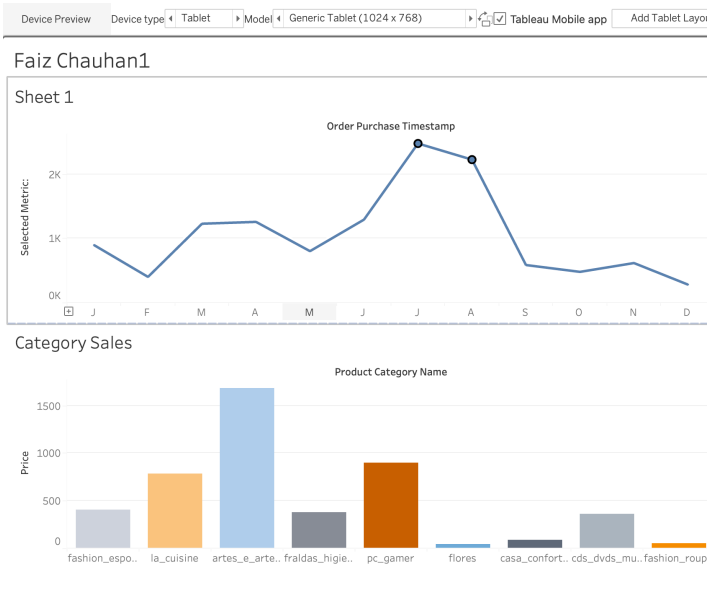


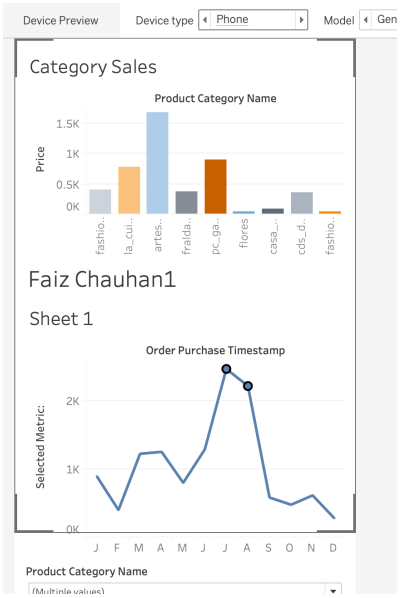


(e) Create dynamic dashboards using parameters

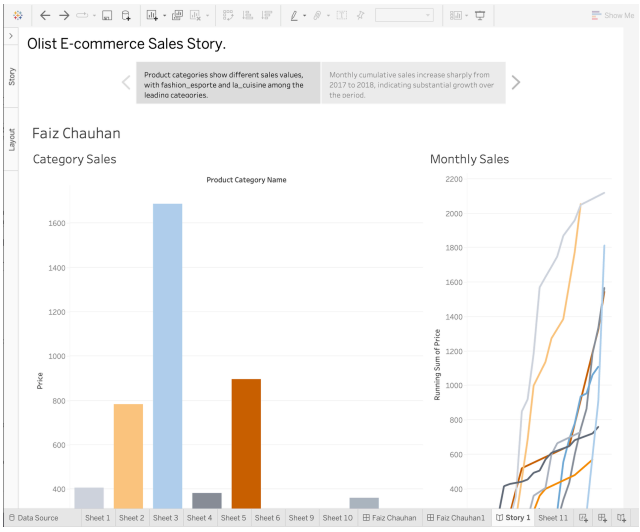


(f) Design dashboards for multiple devices





g) Build a complete data story using Story feature

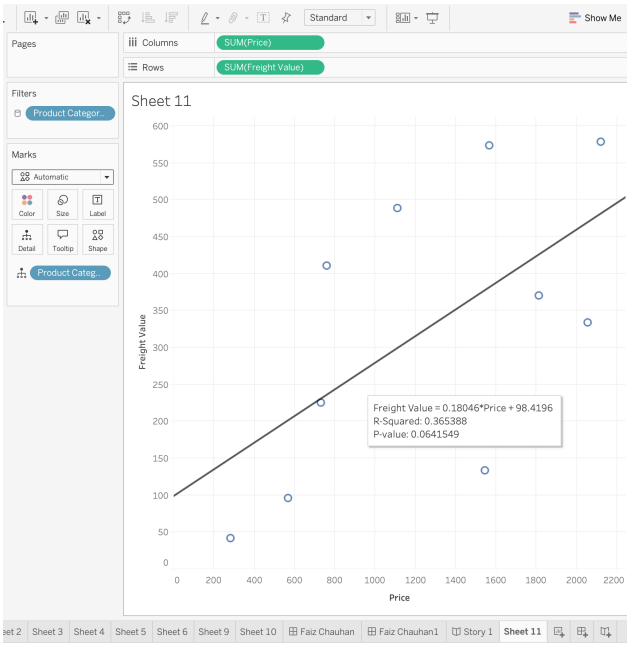


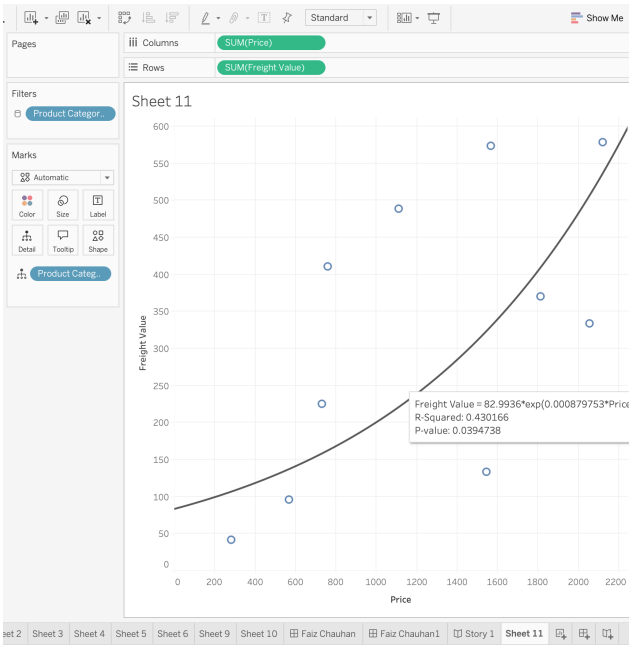
Practical 8

Performing Advanced Analytics using Trends, Clustering, and Forecasting

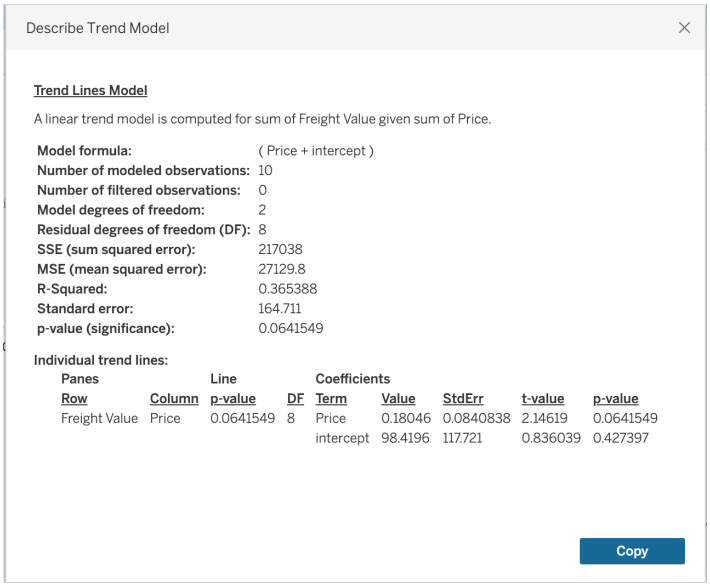
a) Add and interpret trend lines (linear, exponential)

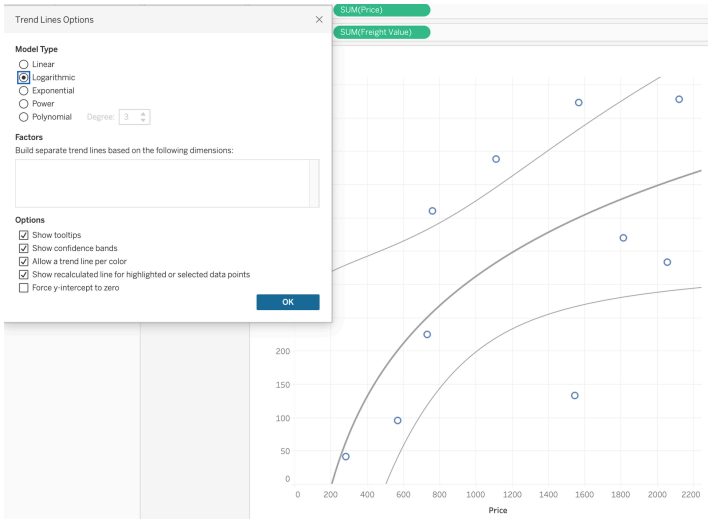
Interpretation: The trend line indicates whether freight value tends to increase as product price increases



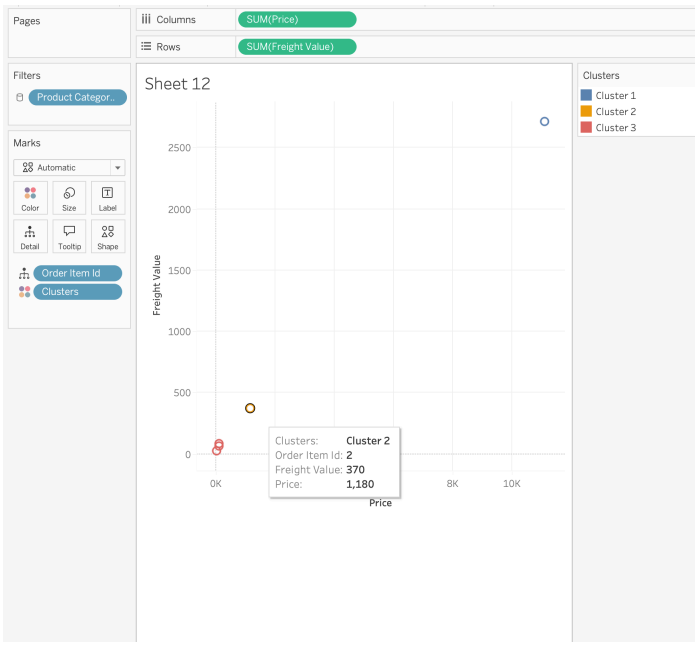


b) Customize and analyze trend models

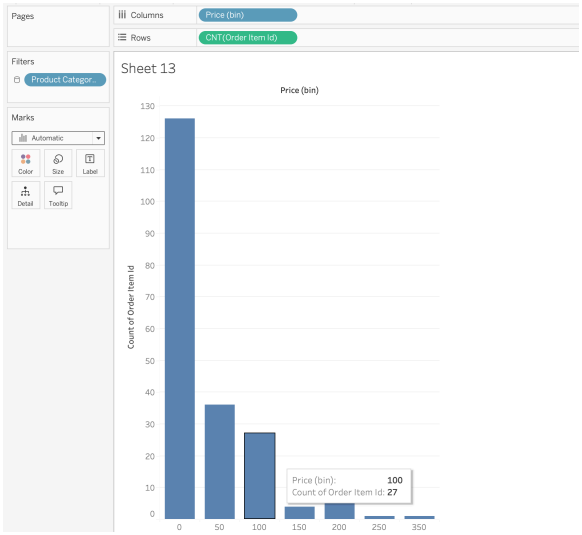




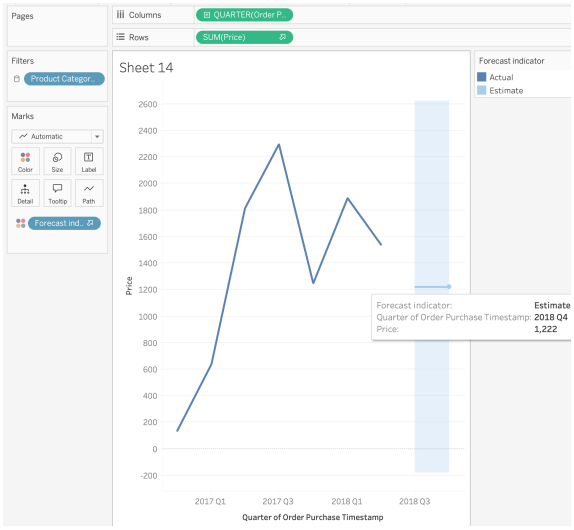
c) Perform clustering on datasets



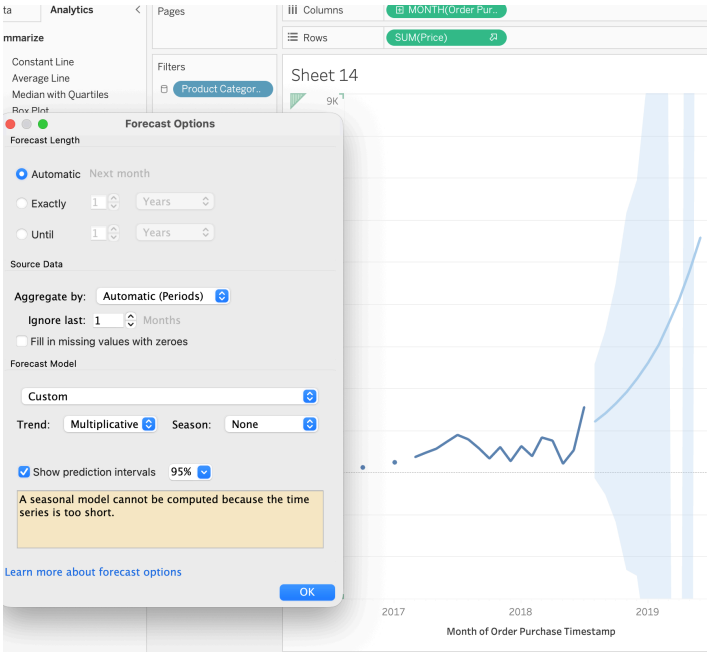
d) Analyze distributions using built-in tools

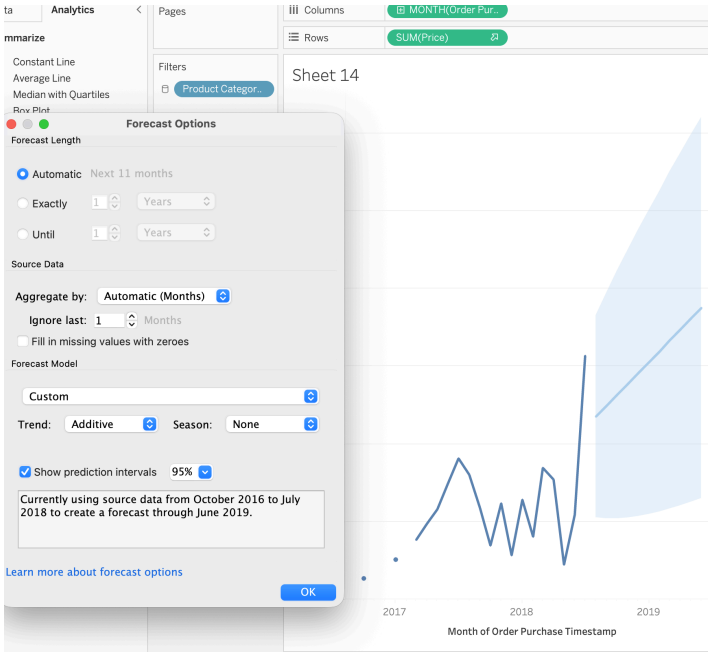


e) Apply forecasting techniques



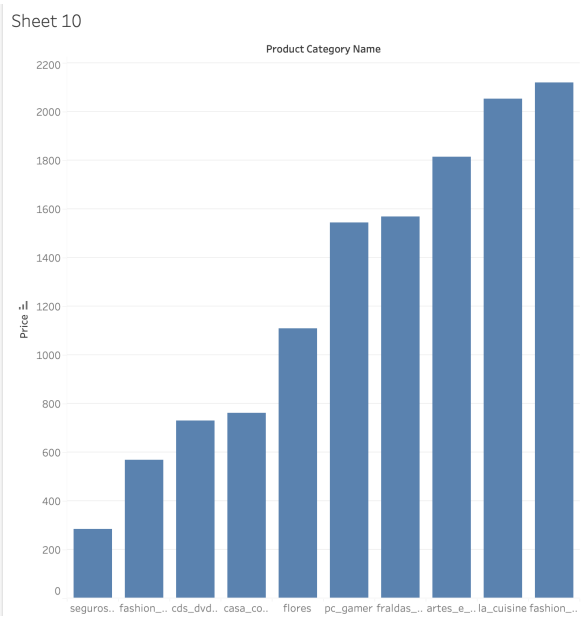
f) Compare different statistical models





g) Export and interpret statistical results

1. **Open a completed chart**
2. **Use Worksheet → Export or the available Download option in Tableau Public.**
3. **Export the visualization as an image or PDF, if supported.**
4. **For underlying data, use View Data and export the displayed data to CSV if that option is available.**

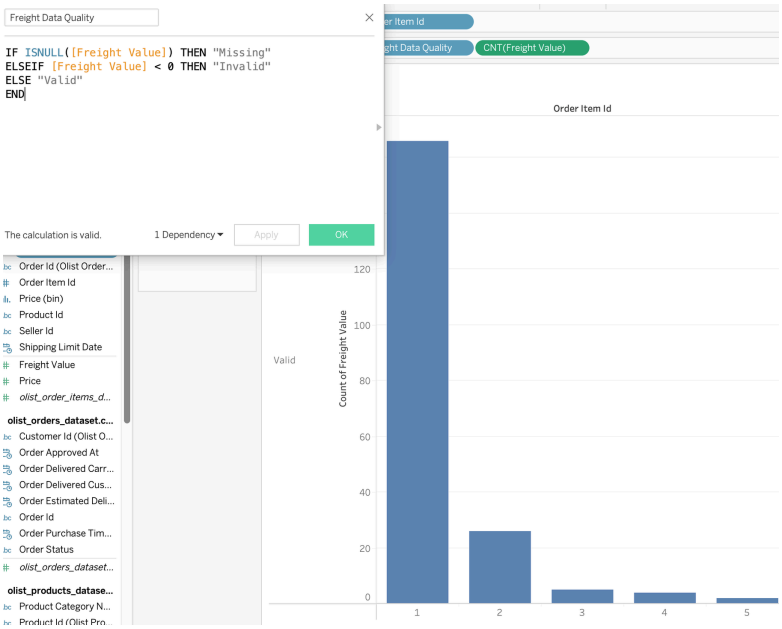


Practical 9

Cleaning and Structuring Data for Analysis

a) Identify and handle missing or inconsistent data

The dataset was examined for missing and inconsistent freight values. No missing values were identified in the field examined. A calculated field was used to classify records as valid, missing, or invalid, supporting the data quality verification process.



b) Convert wide data into tall format

- **Open the Data Source tab.**
- **Connect a suitable spreadsheet or CSV containing multiple columns of similar measurements.**
- **Select the columns you want to restructure.**
- **Right-click the selected column headers and choose Pivot, if available.**

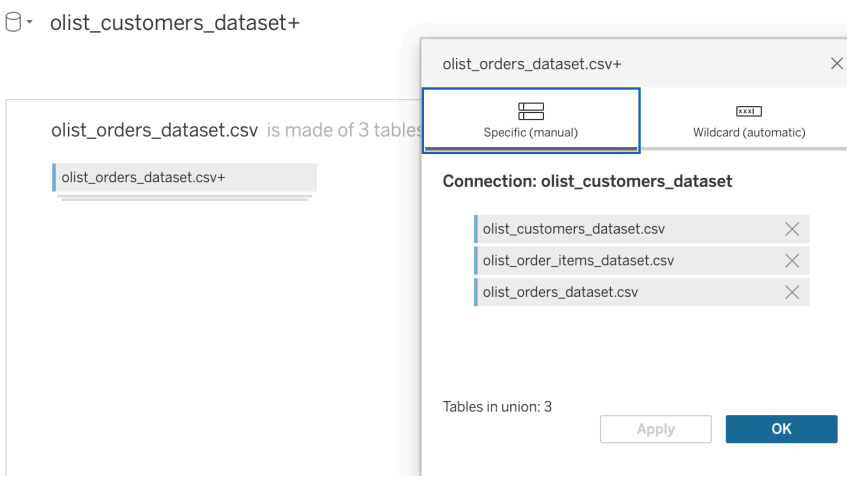
Name

olist_sellers_dataset.csv

Fields

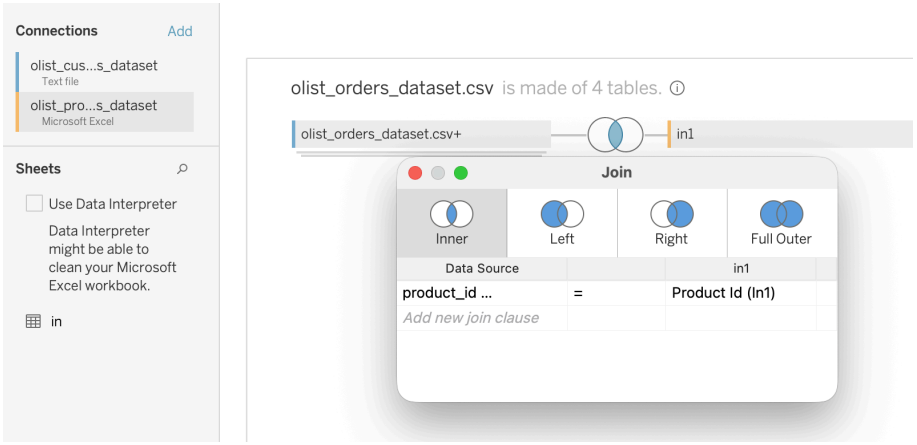
Type	Field Name	Phys...	Rem...
Abc	Attribute	Pivot	Pivot F...
Abc	Pivot Field Values	Pivot	Pivot F...

c) Perform union of multiple files



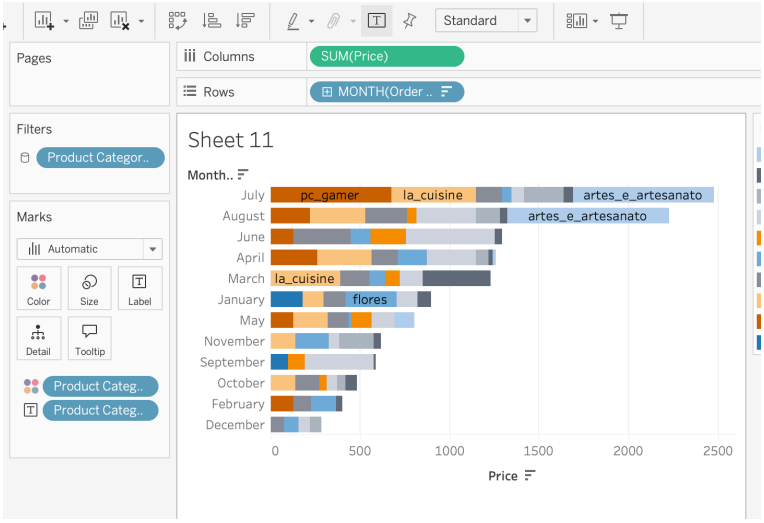
d) Execute cross-database joins

Cross-Database join between an Excel Connection and Text(CSV) Connection using a common key



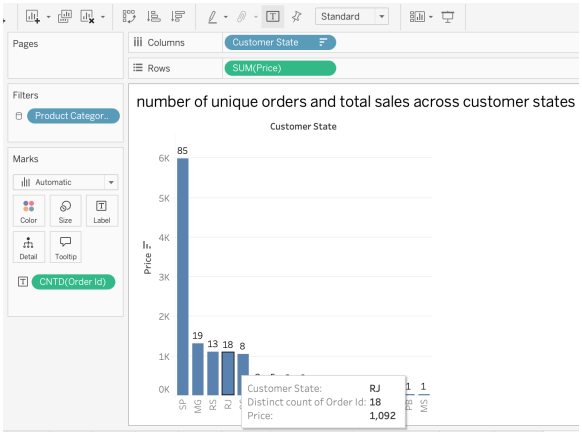
e) Restructure data for better visualization

Sorted by top 10 Categories and added labels for readability



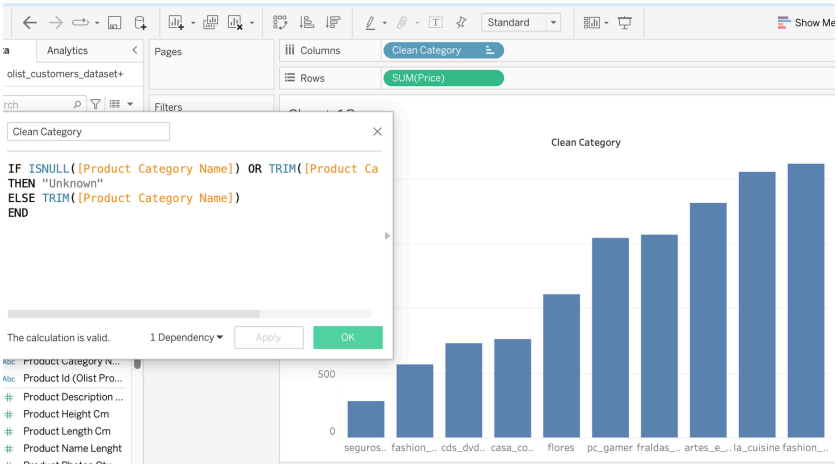
f) Handle different levels of detail in datasets

This compares the number of unique orders and total sales across customer states, showing how data can be summarized at the state level instead of the individual order level.



g) Apply advanced data cleaning techniques

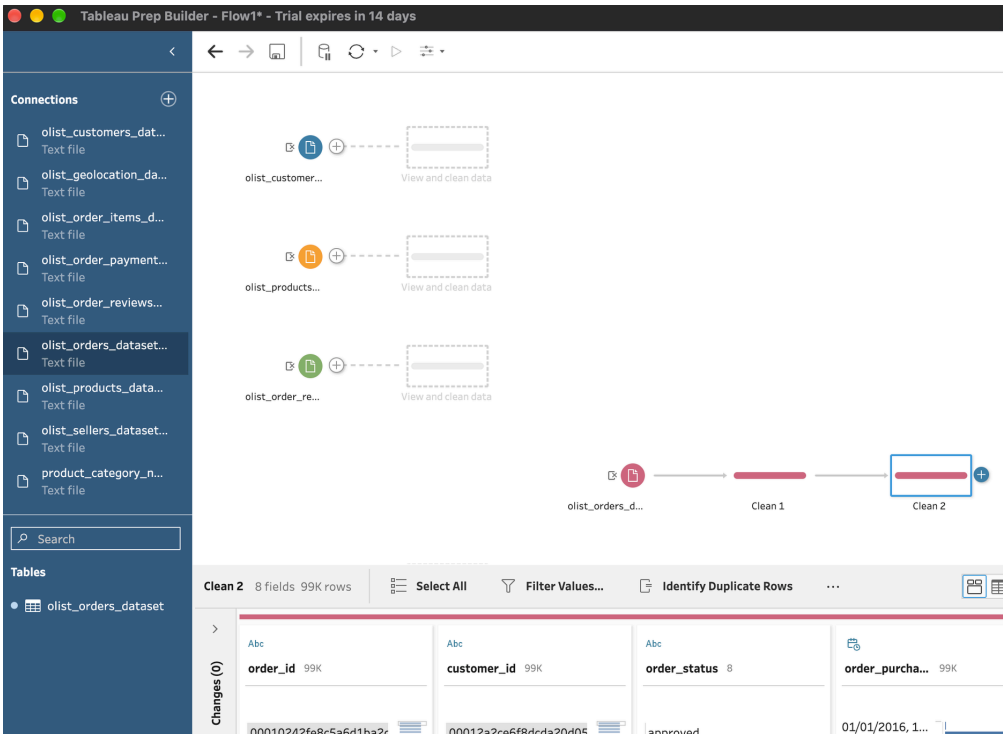
Created a calculated field to replace missing or blank category names with Unknown and remove unnecessary spaces.



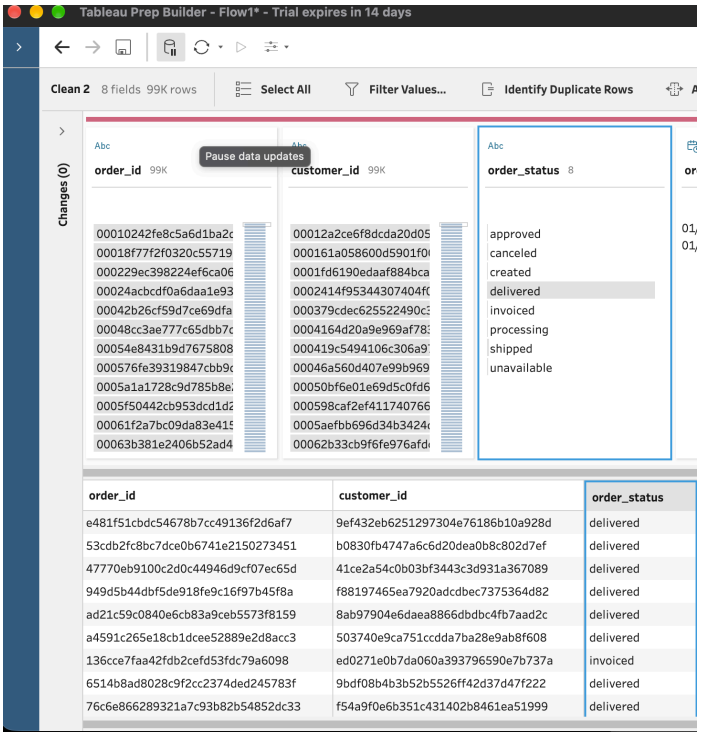
Practical 10

Data Preparation and Transformation using Tableau Prep

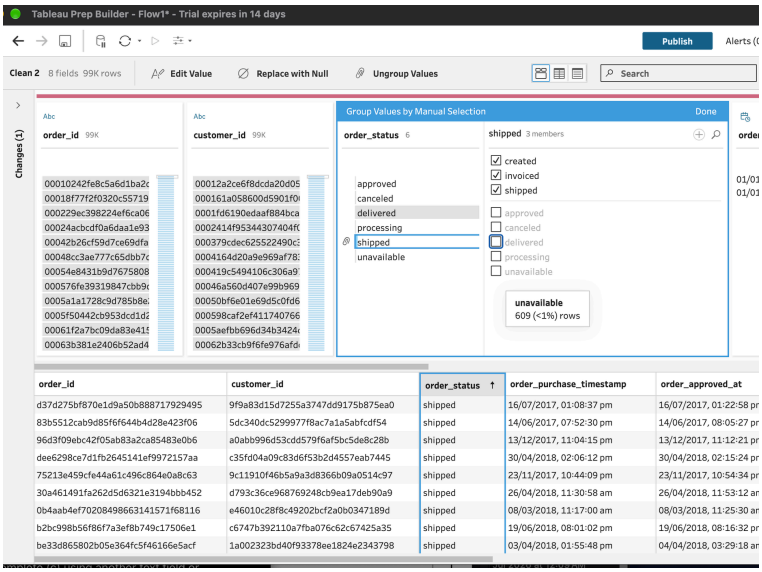
a) Connect and load data into Tableau Prep



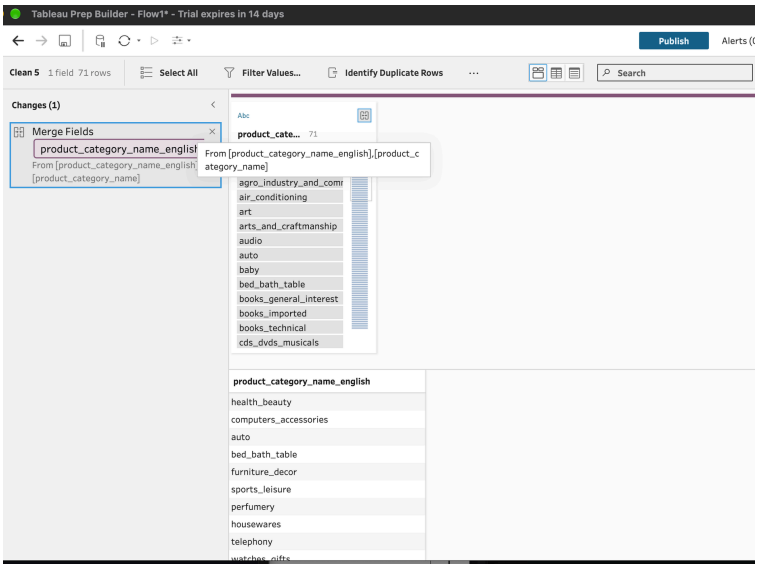
b) Clean and profile data



c) Group and standardize values



d) Merge mismatched fields



e) Perform aggregations in Prep

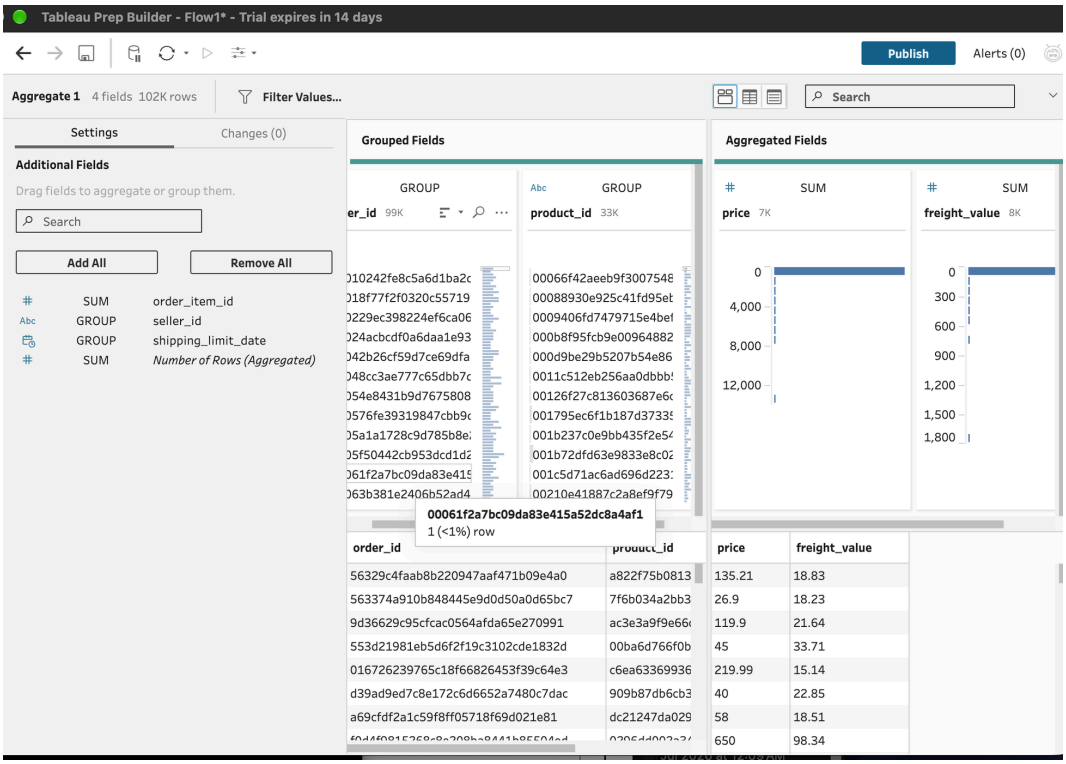
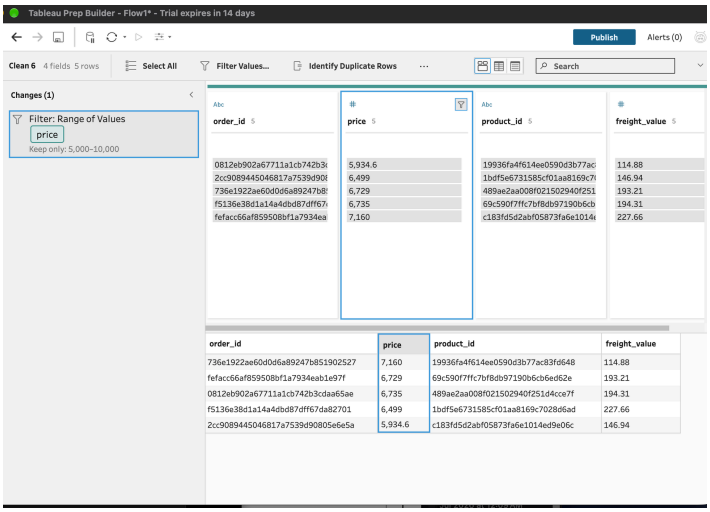
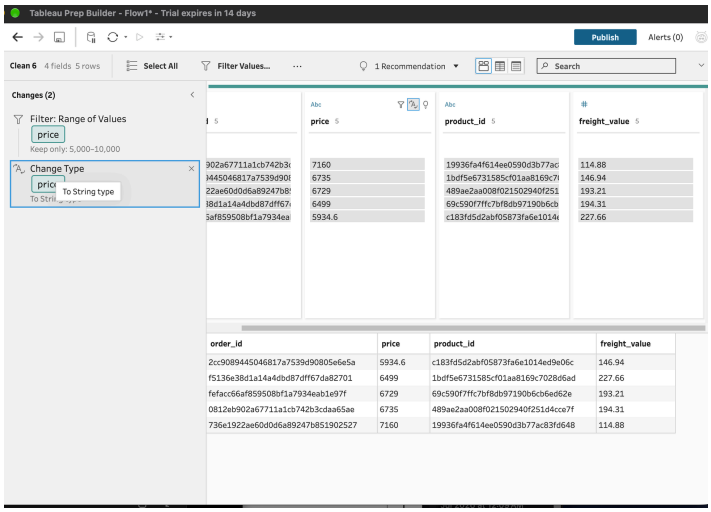


Tableau Prep Builder - Flow1* - Trial expires in 14 days
Publish
Alerts (0)
Aggregate 1 4 fields 102K rows
Filter Values...
Search
Settings
Changes (0)
Additional Fields
Drag fields to aggregate or group them.
Search
Add All
Remove All
SUM order_item_id
GROUP seller_id
GROUP shipping_limit_date
SUM Number of Rows (Aggregated)
Grouped Fields
GROUP
er_id 99K
GROUP
product_id 33K
Aggregated Fields
SUM
price 7K
SUM
freight_value 8K
00061f2a7bc09da83e415a52dc8a4af1
1 (<1%) row
order_id
product_id
price
freight_value
56329c4faab8b220947aaf471b09e4a0
a822f75b0813
135.21
18.83
563374a910b848445e9d0d50a0d65bc7
7f6b034a2bb3
26.9
18.23
9d36629c95cfcac0564afda65e270991
119.9
21.64
553d21981eb5d6f2f19c3102cde1832d
00ba6d766f0b
45
33.71
016726239765c18f66826453f39c64e3
c6ea63369936
219.99
15.14
d39ad9ed7c8e172c6d6652a7480c7dac
909b87db6cb3
40
22.85
a69cfdf2a1c59f8ff05718f69d021e81
dc21247da029
58
18.51
650
98.34

f) Apply filters and transformations



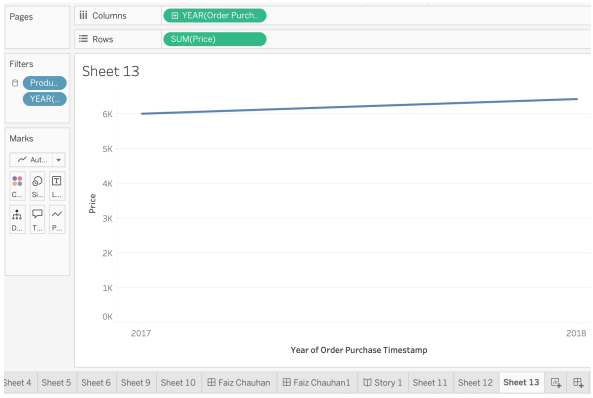


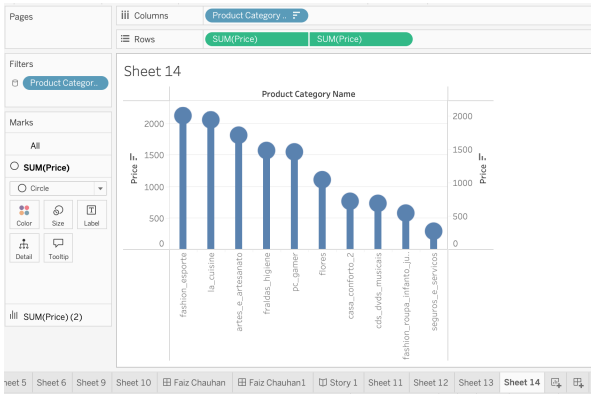
g) Create and automate data preparation flows

Practical 11

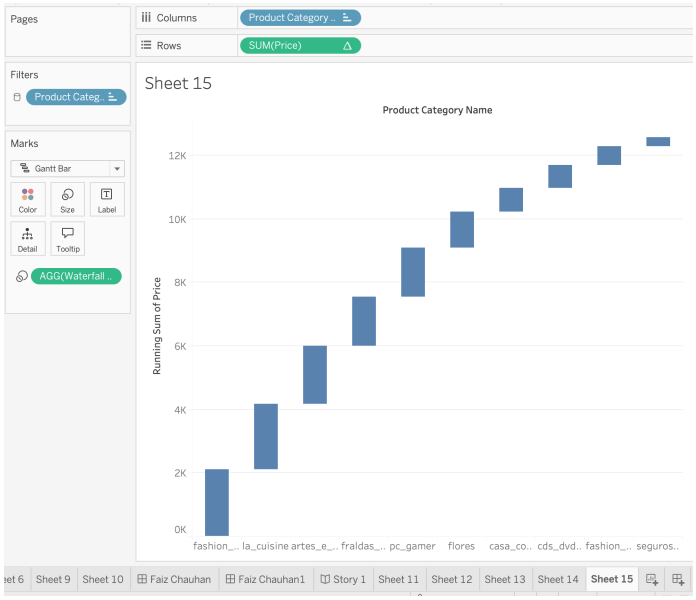
Implementing Advanced Visualization Techniques and Customizations

a) Create slope charts and lollipop charts

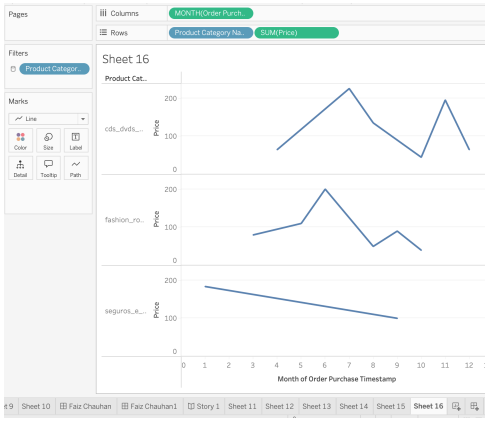




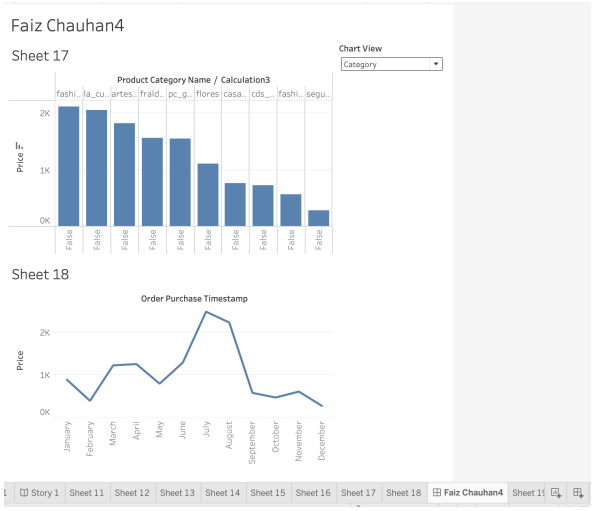
b) Design waterfall charts

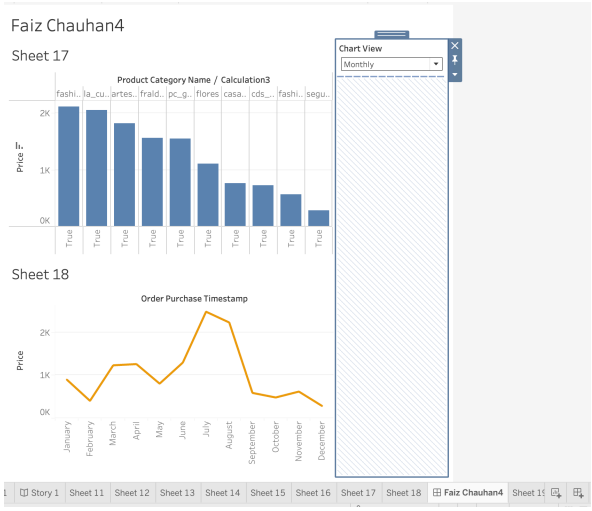


c) Build sparklines for trend analysis

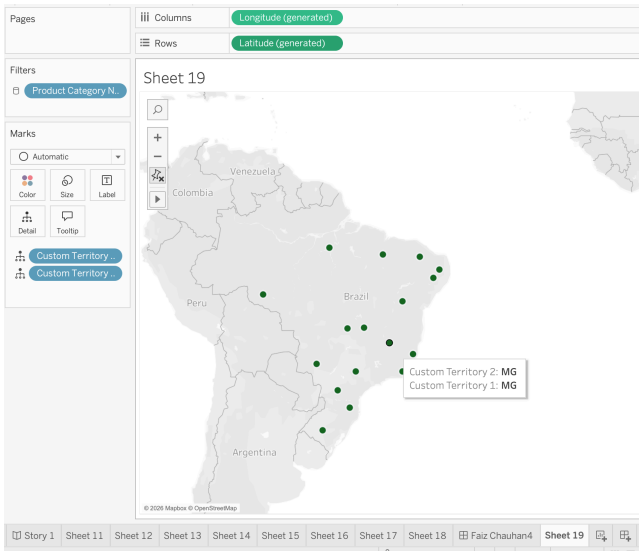


d) Implement sheet swapping for dynamic dashboards

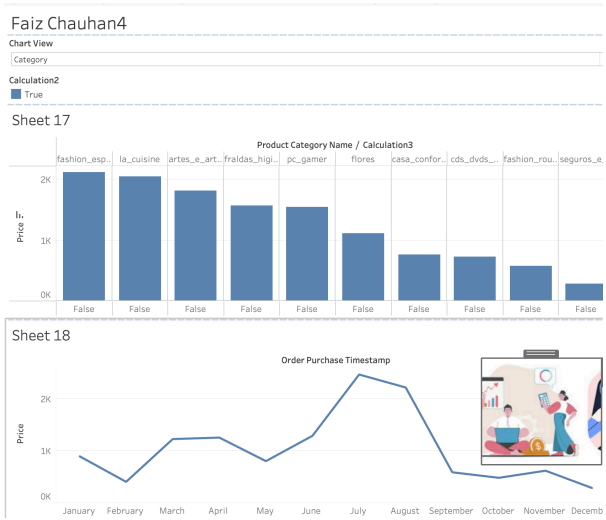




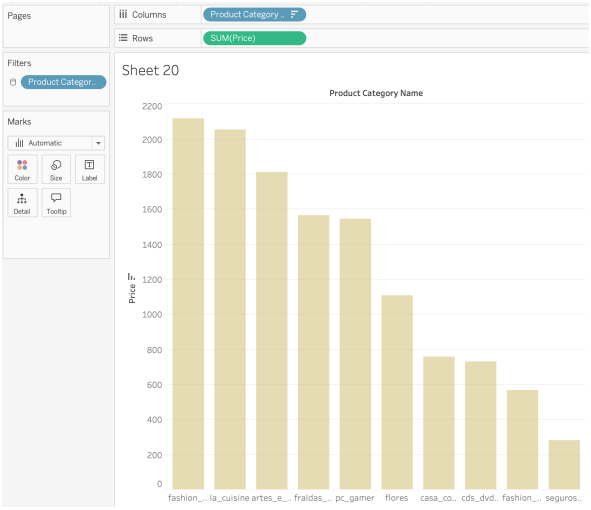
e) Create custom geographic territories

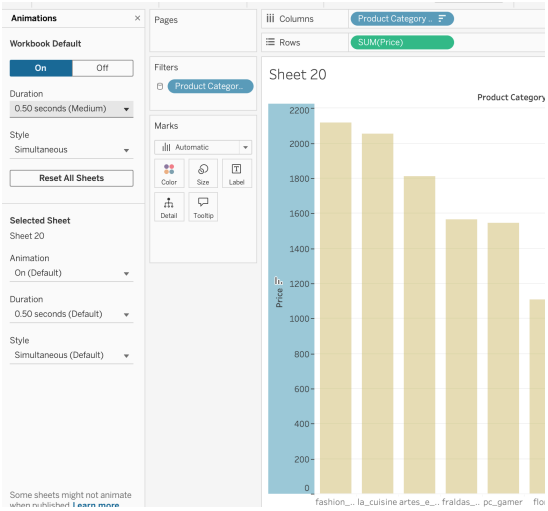


f) Use background images in dashboards



g) Apply animation and transparency techniques

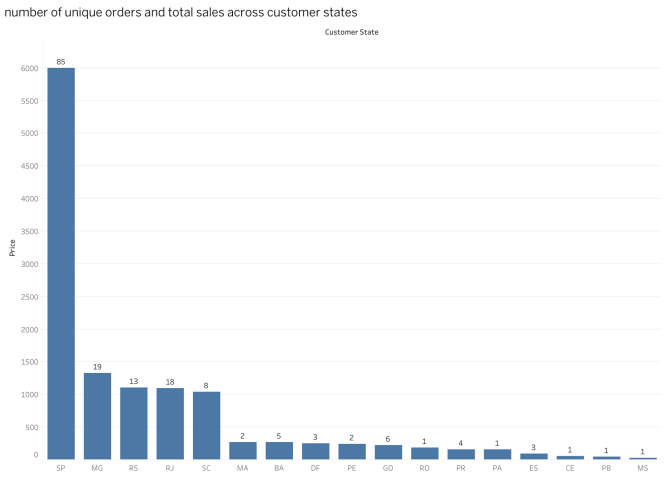




Practical 12

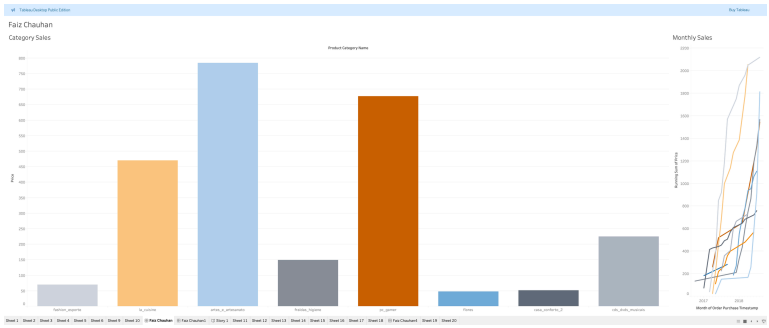
Sharing, Publishing, and Presenting Data Visualizations

a) Export dashboards as PDF/Image

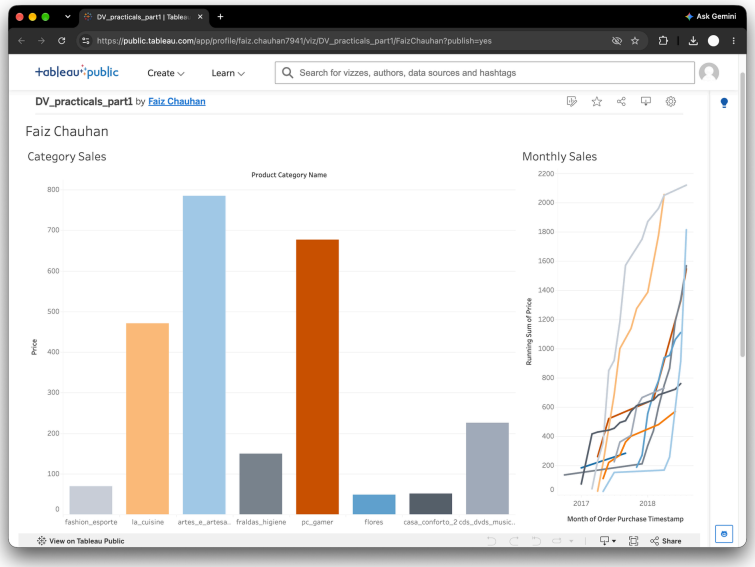


- **Open your dashboard on the Tableau Public Website.**
- **Click the Download button located on the top or bottom toolbar of the visualization frame.**
- **Select Image from the menu options.**
- **The dashboard will download as a PNG image to your computer's default downloads folde**

b) Use presentation mode in Tableau



c) Publish dashboards to Tableau Public

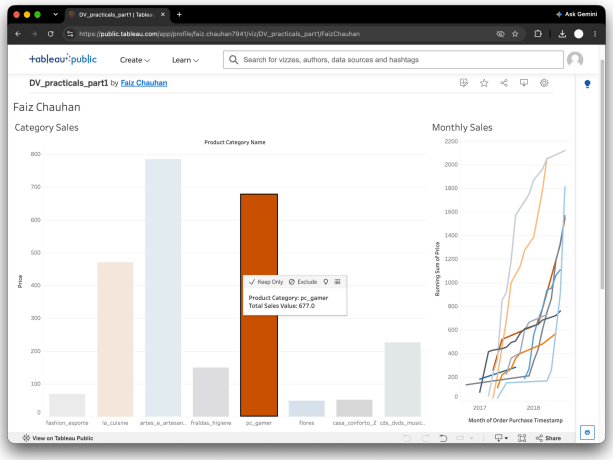


d) Share dashboards via Tableau Server/Online

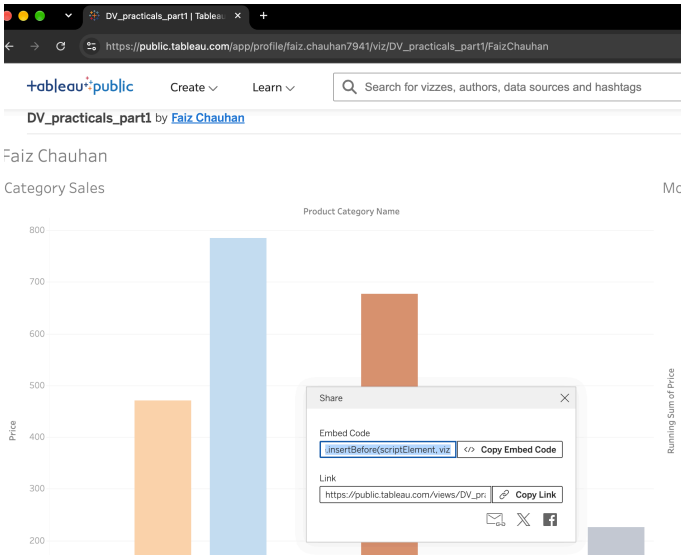
Share dashboards via Tableau Server/Online

1. **Open your published dashboard in a browser.**
2. **Copy the URL from the address bar.**

e) Enable user interaction on shared dashboards



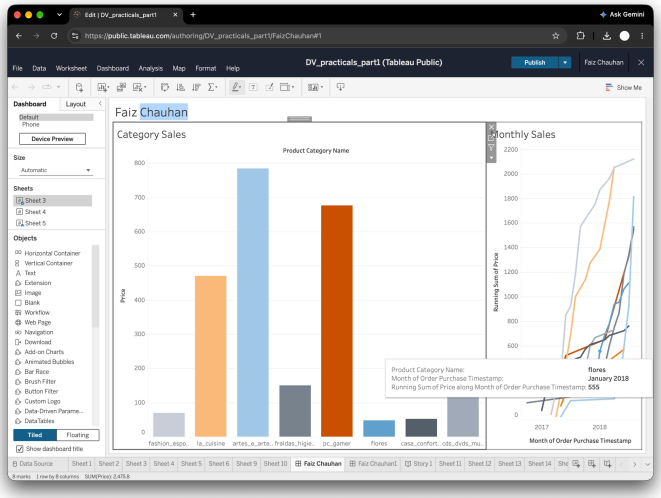
f) Manage permissions and access control



g) Evaluate and optimize sharing strategies

Steps

1. **Open your completed dashboard in Tableau.**
2. **Make sure the title is clear.**
3. **Check that chart titles, axis labels, and category names are readable.**
4. **Remove any unnecessary blank space, legends, or objects.**
5. **Test the Chart View parameter and any filters to confirm they work.**
6. **Open the dashboard in Tableau Public in your browser, if published, and check that it displays correctly.**
7. **Resize the browser window to check readability at a smaller screen size.**



Practical 13

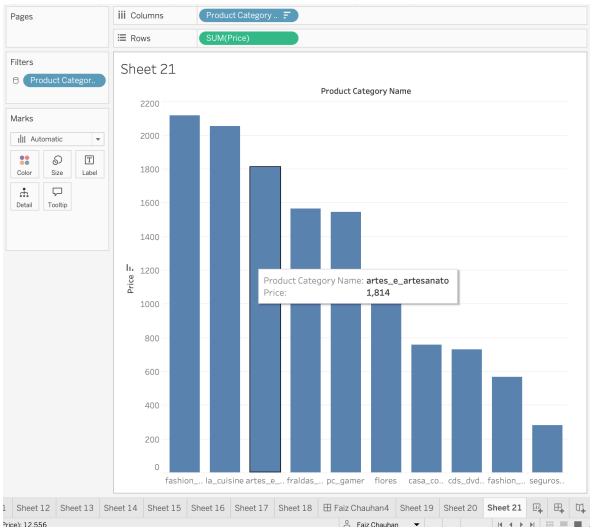
Retail Sales & Profitability Dashboard

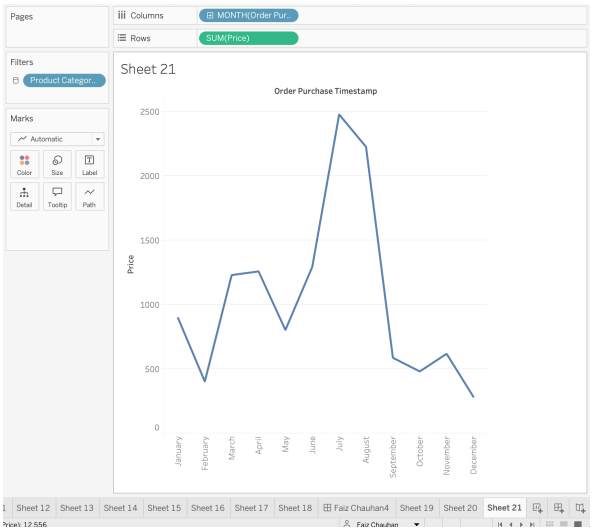
Objective: Analyze sales, profit, and discount impact across regions.

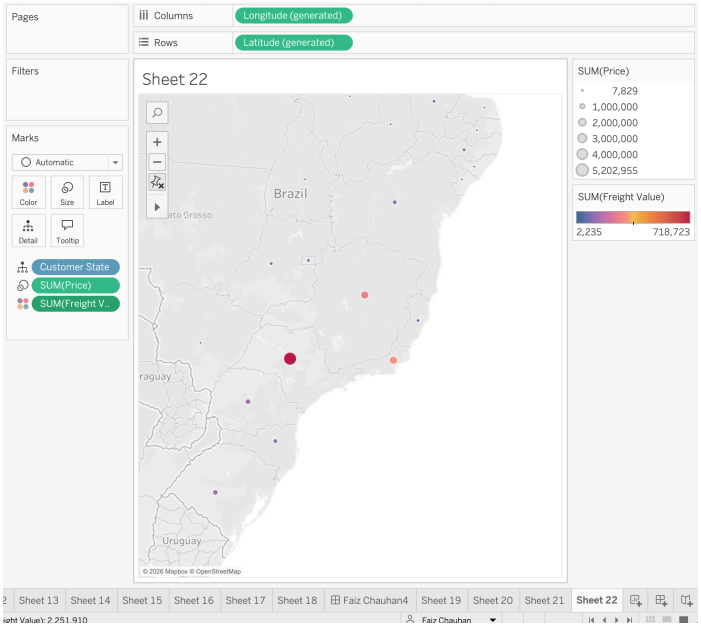
Key Tasks:

1. Build bar, line, and map visualizations
2. Use calculated fields (Profit Ratio)
3. Apply filters (Region, Category)
4. Create interactive dashboard

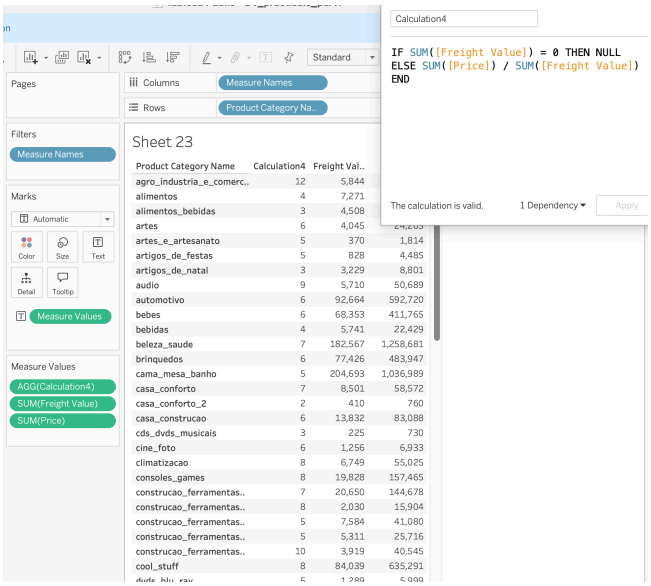
1.Build bar, line, and map visualizations



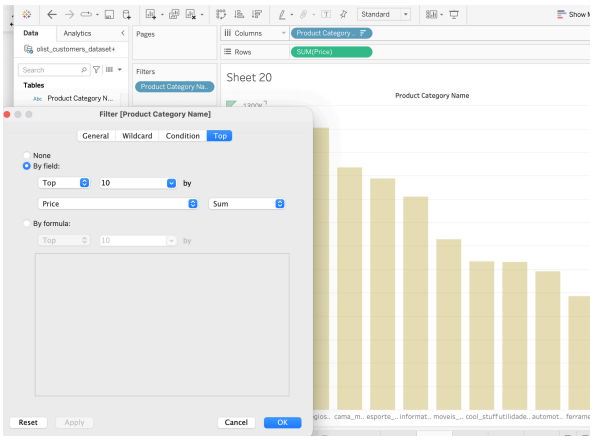


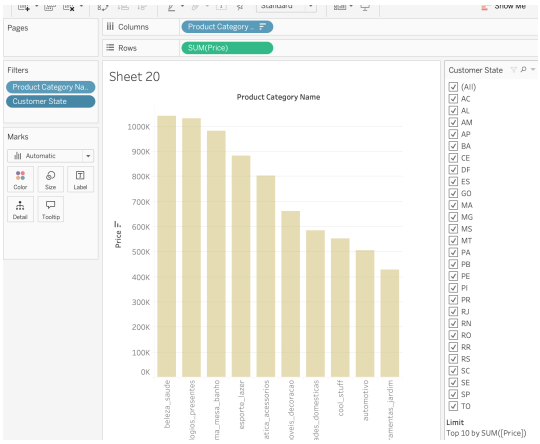


2.Use calculated fields (Profit Ratio)

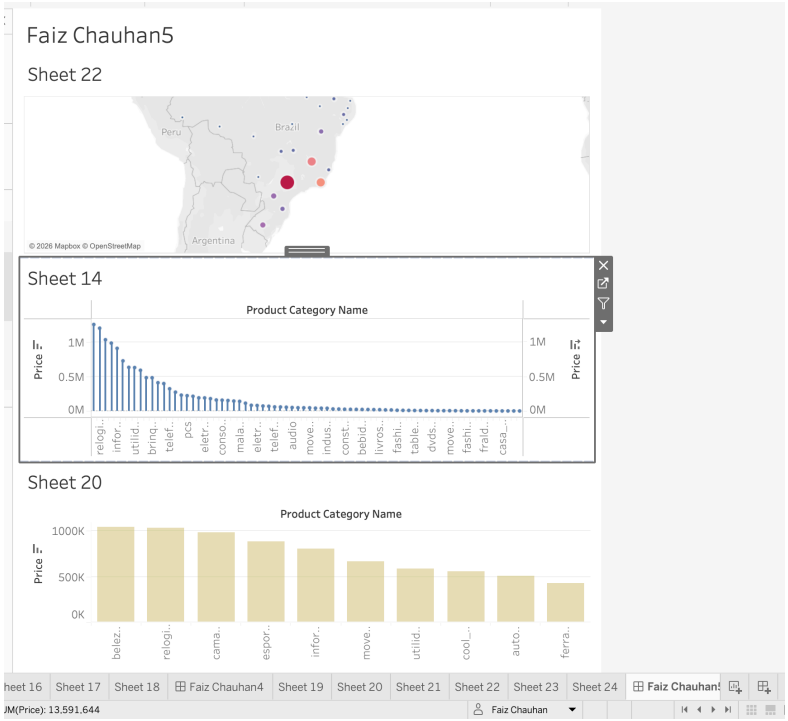


3.Apply filters(Region,Category)





4.Create Interactive Dashboard



Practical 14

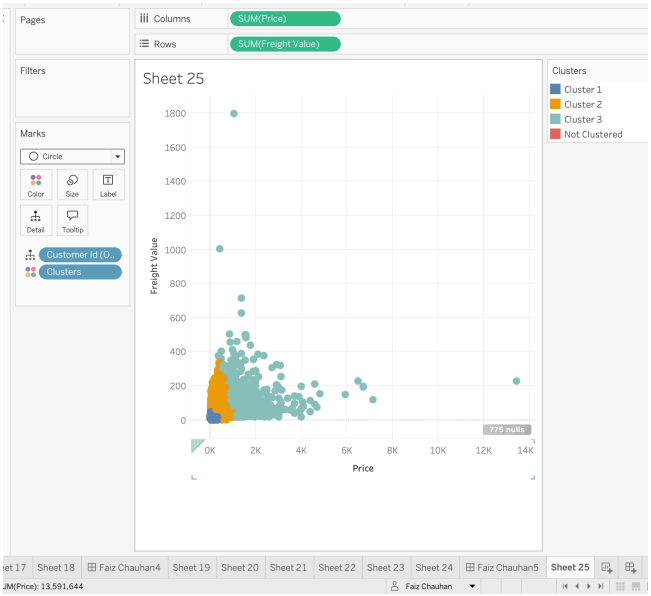
Customer Segmentation & Behavior Analysis

Objective: Identify customer segments based on purchasing patterns.

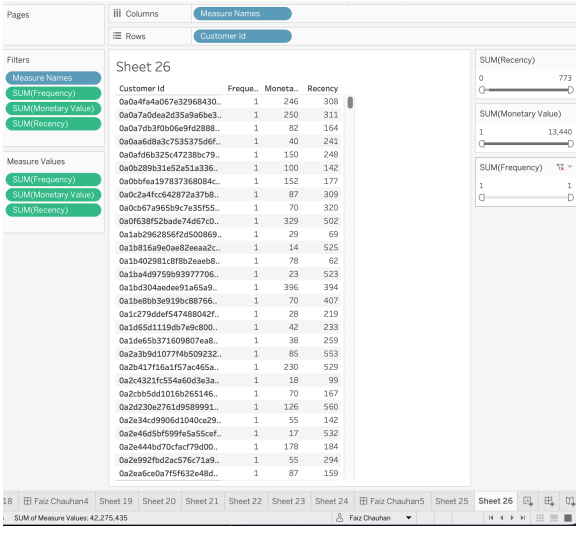
Key Tasks:

- Use clustering
- Analyze frequency, recency, and monetary value
- Create scatter plots and segmentation dashboards
- Highlight high-value customers

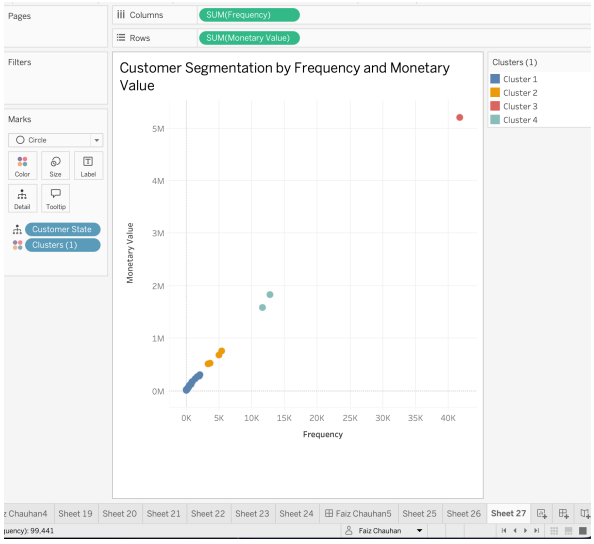
1.Use Clustering

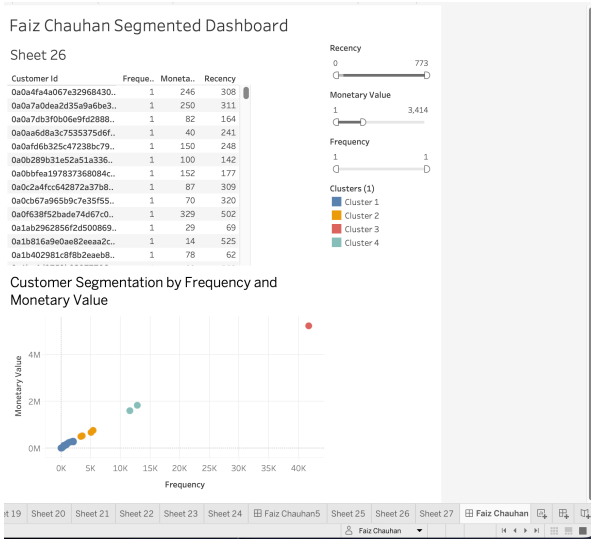


2.Analyze frequency, recency, and monetary value (RFM)

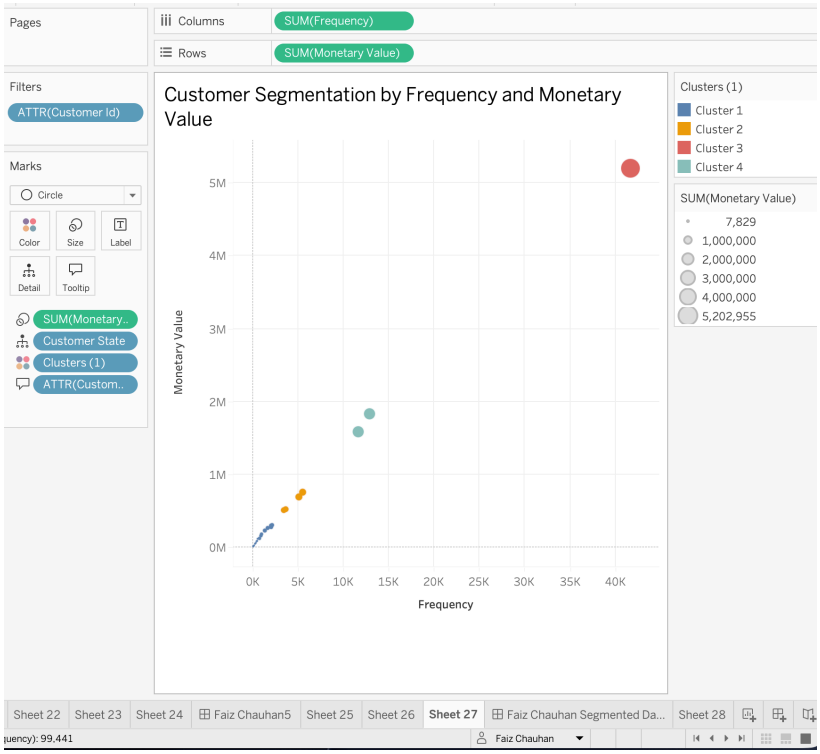


3.Create scatter plots and segmentation dashboards





4.Highlight High value customers



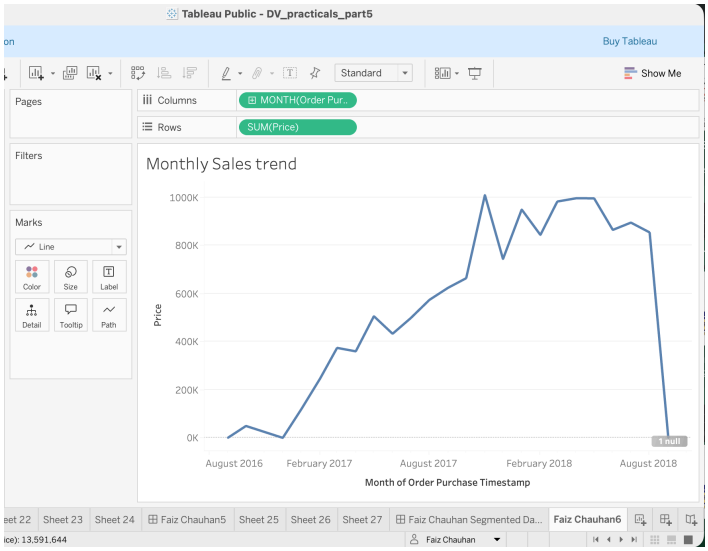
Practical 15

Objective: Predict future sales using historical data.

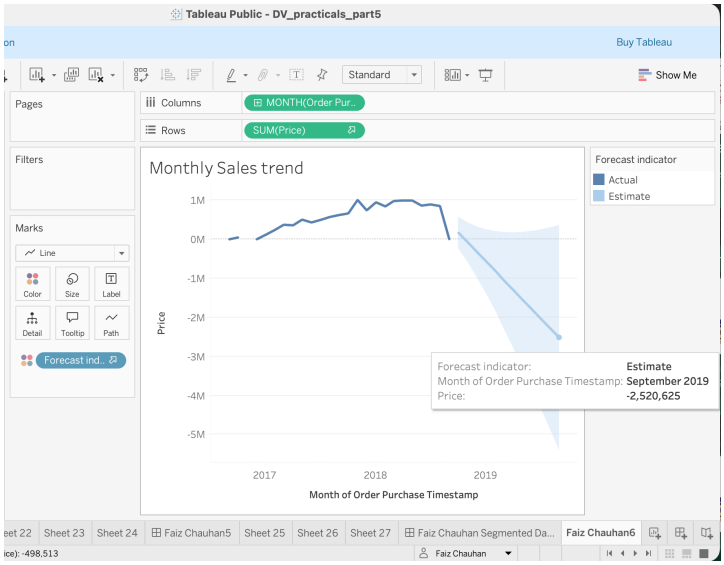
Key Tasks:

- Build time-series charts
- Apply forecasting
- Compare seasonal trends
- Use moving averag
- Use moving averages

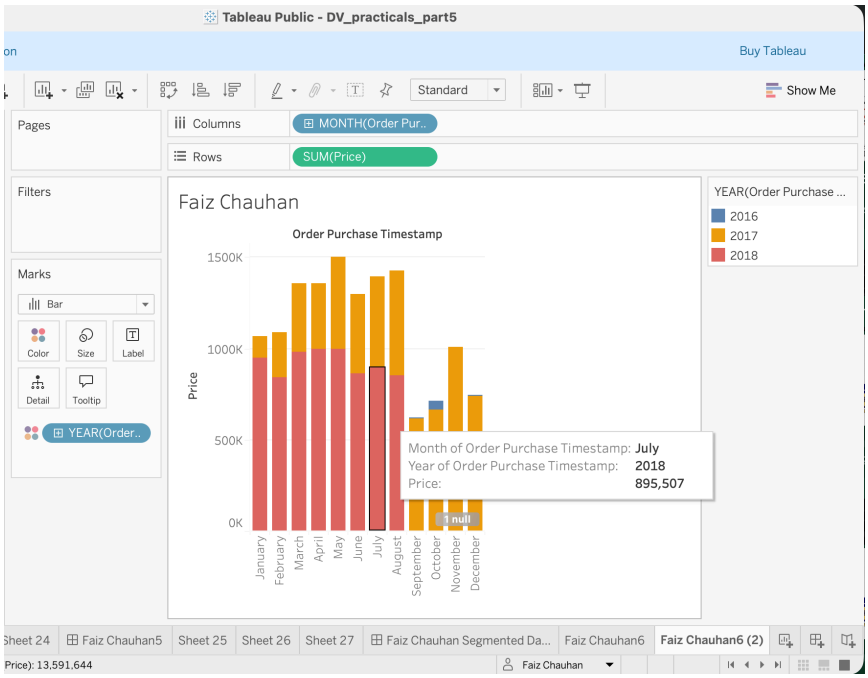
Building time series charts



Apply Forecasting



Compare seasonal trends



Use moving Averages